

Seabed mobility study of Site N-3.8 in Nordseecluster Offshore Windfarm



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Client	Nordseecluster B GmbH
Contact	Ben de Sonnevile
Reference	NSC_PR_2022-0084
Keywords	Seabed mobility, morphodynamics, Nordseecluster

Document control

Version	1.0
Date	21-04-2023
Project nr.	11208993-002
Document ID	11208993-002-HYE-0002
Pages	50
Classification	
Status	final

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Doc. version	Author	Reviewer	Approver
1.0	Vassia Dagalaki	Tom Roetert 	Paul van Steeg

Summary

The Nordseecluster Offshore Wind Farm (OWF) is a proposed development located in the North Sea, approximately 55 km off the German coast. The project is being developed by Nordsee Two GmbH and Nordseecluster B GmbH and consists of four sites (N-3.5, N-3.6, N-3.7 and N-3.8), auctioned as standalone sites, but developed as a single combined project. This report describes the methods and results of the seabed mobility analysis within and around one of the sites, Site N-3.8. The objectives of this seabed mobility study are to assess the morphodynamics and provide an estimation of the future seabed levels in the area of interest. To this end, data-driven analysis is performed that consists of five parts: overview of the characteristics in Site N-3.8, morphodynamic analysis, estimation of future bed levels, Unexploded Ordnance (UXO) analysis and a description of possible risks and mitigation measures.

The German Bight is characterised as a meso-tidal to macro-tidal environment. Within Site N-3.8 current velocities are typically along the east-west axis, ranging between 0 to 1.6 m/s. Current velocities are typically smaller than 0.5 m/s. Significant wave heights in Site N-3.8 are generally smaller than 2.9 m. Residual sediment transports are westwards due to the effect of prevailing tide and meteorological conditions. The seabed in Site N-3.8 can be characterised as a sloped seabed with locally minor undulations. Bed levels in the area range from -27 to -34 m relative to the LAT. Rhythmic bedforms are observed in the most recent survey for Site N-3.8 with heights ranging up to 0.10 m. The surficial sediment in and around Site N-3.8 is predominantly classified as fine sand with a decreasing median diameter towards the northeast. The layer consists of Holocene marine sand mixtures (silty sand), overlying layers of fine to coarse sand. Occasionally layers of peat or low strength clay are observed close to the seabed surface (within four metres below the seabed).

The morphodynamic analysis showed that over the period 1992 to 2022 the mobility of the seabed in Site N-3.8 is limited. Nevertheless, seabed mobility cannot be fully excluded due to the bathymetric surveys' limited temporal and spatial coverage and indications of survey inaccuracy of the bathymetric datasets. After correcting the bathymetries for potential errors in the vertical referencing, the remaining maximum seabed changes are 0.12 m over a period of 4 years and 0.46 m over a period of 26 years. The spatial distribution of the remaining seabed changes suggests that the most extreme values might still be attributed to survey inaccuracies/errors rather than a seabed change trend. The analysis showed no clear indications of a persistent trend in seabed level change.

To further delineate the origin of the observed bed level variability, the analysis was expanded to include datasets of neighbouring sites as well as data- or numerical model- derived estimates of morphodynamic activity available in literature. The remaining maximum seabed changes within Site N-3.7 are 0.14 m over a period of 3 years and 0.20 m over a period of 27 years. Similar to Site N-3.8, for this area there are no clear indications of a persistent seabed level change trend. The recorded bed level changes in the neighbouring Nordsee One OWF show alternating patterns of erosion/ accretion (within ± 0.25 m) over different parts of the windfarm and different periods. Data driven estimates of morphodynamic activity of the surrounding areas within the German and Dutch continental shelf show limited morphodynamic activity, in the order of 0.25 m over 14 to 26 years. Lastly, morphodynamic simulations indicate an overall long-term accretive behaviour of maximum 0.5 m/100 years.

Estimation of future bed levels accounts for seabed changes caused by small-scale dynamics (i.e., migration of megaripples) and large-scale dynamics (i.e., regional seabed changes).

To account for this, we consider a uniform variability range of ± 0.25 m over the full extent of Site N-3.8. This range accounts for the migration of megaripples such as those observed and bed level changes similar to the differences observed in the data and literature. Together with the uncertainties related to multibeam measurement (± 0.20 m, (International Hydrographic Organization, 2020)) a maximum change of ± 0.45 m with respect to the smoothed BSH 2018 bathymetry is expected at any given location over the lifetime of the windfarm (30 years, until 2053).

The lowest and highest expected levels where UXOs might have migrated to during the period 1945-2018 were estimated based on the results of the morphodynamic analysis, -0.60 m and $+0.2$ m from the smoothed BSH 2018 bathymetry respectively. Finally, risks related to the predicted morphodynamic changes and possible mitigation measures are described.

The predicted seabed level changes presented in this study follow from the applied morphological analysis techniques, describing the (uncertainty of the) physics and the natural variability of the analysed morphological system. No additional safety margins for design purposes have been applied.

Although it is concluded that there is little potential for seabed mobility in this region, there was limited availability of data to confirm this further. It is thus recommended to further expand the current available bathymetry datasets over time (during installation), to confirm the anticipated behaviour as described in this report.

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1 Introduction

1.1 Background

The Nordseecluster Offshore Wind Farm (OWF) is a proposed development located in the North Sea, approximately 55 km off the German coast, see Figure 1.1. The project is being developed by Nordsee Two GmbH and Nordseecluster B GmbH (hereby referred to as the Client). The project consists of four sites (N-3.8, N-3.7, N-3.6 and N-3.5), auctioned as standalone sites, but developed as a single combined project. Insight into the mobility of the seabed in and around the Nordseecluster OWF Sites is important to determine if risks can be expected in the supply of generated wind energy, e.g., with respect to the stability of wind turbines and the exposure of power cables.

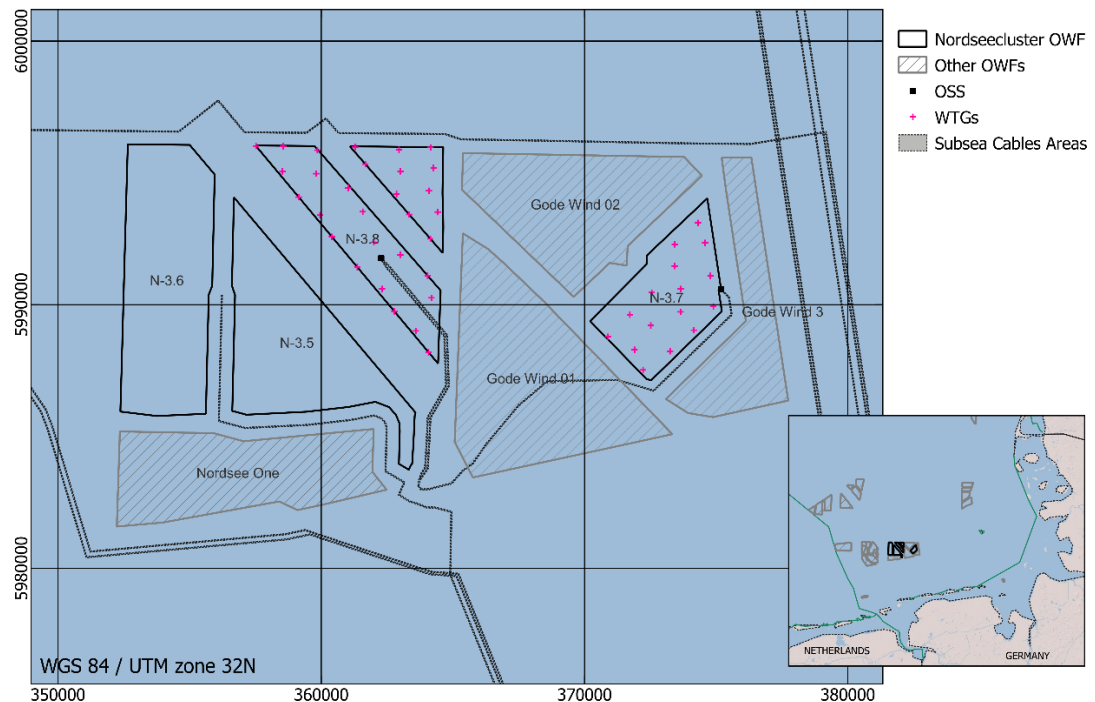


Figure 1.1 Map showing the boundaries of Nordseecluster OWF (black lines), Wind Turbine Generator (WTG) locations (magenta cross), Offshore SubStation locations (black rectangles), subsea cable areas (grey hatch) and other Offshore Wind Farm Sites (grey line hatch). The lower right panel shows the Nordseecluster OWF on a larger scale.

1.2 Objectives

The overall objective of this seabed mobility study is to assess the morphodynamics in and around site N-3.8 of Nordseecluster OWF (see Figure 1.1). Site N-3.8 consists of two separate areas in the shapes of a trapezoid (approx. 16 km²) and a triangle (approx. 7 km²). Furthermore, the objective is to provide an estimation of the future seabed levels in the area for a period of 30 years (2023-2053). Additionally, the impact of morphodynamics on possible burial and migration of Unexploded Ordnances (UXOs) is to be assessed. Finally, risks related to the predicted morphodynamic changes and possible mitigation measures are to be described.

1.3 Methodology

To achieve the above-mentioned objectives, a data-driven analysis is performed. The inputs, methodology, results and conclusions of this analysis are presented in the next chapters.

Firstly, the available data is processed and reviewed in Chapter 2. An overview of the hydrodynamic, wave, bathymetric, geotechnical and geophysical characteristics in and around Site N-3.8 is presented.

The data-driven analysis of site N-3.8 is presented in Chapter 3 and consists of five parts: overview of the seabed characteristics, morphodynamic analysis, estimation of future bed levels, Unexploded Ordnance (UXO) analysis and a description of possible risks and mitigation measures.

The morphodynamic analysis will demonstrate to what extent (if present) bed features such as mega ripples, sand waves and sand banks are present in and around Site N-3.8. It is important to make this characterisation since these features change and migrate under different circumstances, hence determine the extent of the mobility of the seabed. Moreover, multiple bathymetry datasets are used to determine if and where the bathymetry changes occur. Thereafter, the observed bed features and bed level changes are validated using the available hydrodynamic, wave, geotechnical and geophysical data.

The results of the morphodynamic analysis are used to determine morphodynamic trends and uncertainties. Those trends and uncertainties are used to make an estimation of the future bed levels in and around Site N-3.8 of Nordseecluster OWF. Additionally, the derived morphodynamic trends are used to qualitatively assess the most likely depths for encountering (buried) UXOs. Finally, relative risks and possible mitigation measures regarding the calculated seabed level changes along with mitigation/avoidance measures are presented.

The study is concluded with the conclusions and recommendations, outlined in Chapter 4.

2 Overview of available data

2.1 Introduction

This section contains an overview of the available data. The available bathymetric datasets are described in Section 2.2. Section 2.3 outlines the geophysical and geotechnical data. The hydrodynamic and wave data is presented in Section 2.4. Other data used in the analysis are outlined in Section 2.5.

2.2 Bathymetric data

The two main sources of bathymetric data for this study comprise of (i) one existing 'historical' bathymetrical dataset available from the client and (ii) two existing 'historical' dataset available from BSH (the Federal Maritime and Hydrographic Agency of Germany) (Bundesamt für Seeschifffahrt und Hydrographie (BSH), 2022, Ramboll, 2022). These bathymetries are used to determine the seabed characteristics and to estimate future seabed levels in and around Site N-3.8.

Table 2.1 contains detailed information on the three surveys, including measurement dates, vertical reference level, data coverage and resolution.

Table 2.1 Historical bathymetrical surveys available in and around Site N3-8.

Survey ID	Survey Name	Survey date [mm-yyyy]	Vertical reference level	Data coverage	Resolution [m]
Site specific bathymetry data – received from the client					
N0305_N0306_N0308_25832_LAT_1m (Ramboll, 2022)	BSH N-3.8, 2018 bathymetry	09-2018	LAT	Full coverage of the site, N-3.5 & N-3.6	1
0.2m5p_bathy160422PM_Final Offshore	N-3.8 UXO survey 2022	04-2022	LAT	ca. 1600 x 70 m boxes at several locations	0.20
Bathymetry data for wider area – obtained from various sources					
Deutsche Bucht 3, interpolated (Bundesamt für Seeschifffahrt und Hydrographie (BSH), 2022)	Deutsche Bucht 3	1992	LAT	German EEZ	200 - 500

Figure 2.1 shows the bathymetries of the two surveys conducted in 2018 and 2022 (i.e., BSH N-3.8 2018 bathymetry & N-3.8 UXO survey 2022). Finally, Figure 2.2 shows the 1992 interpolated Deutsche Bucht 3 bathymetry.

Other data sources such as GEBCO and data from admiralty charts are not used in this morphodynamic analysis because of limited spatial resolution and coverage and large uncertainties in bed levels. Bathymetry DTMs from EMODnet and EasyGSH are used only for the visualisation of the bathymetry of the German Bight. They are however not used in comparison with the surveys listed in Table 2.1 as they do not consist of original data in the area of interest.

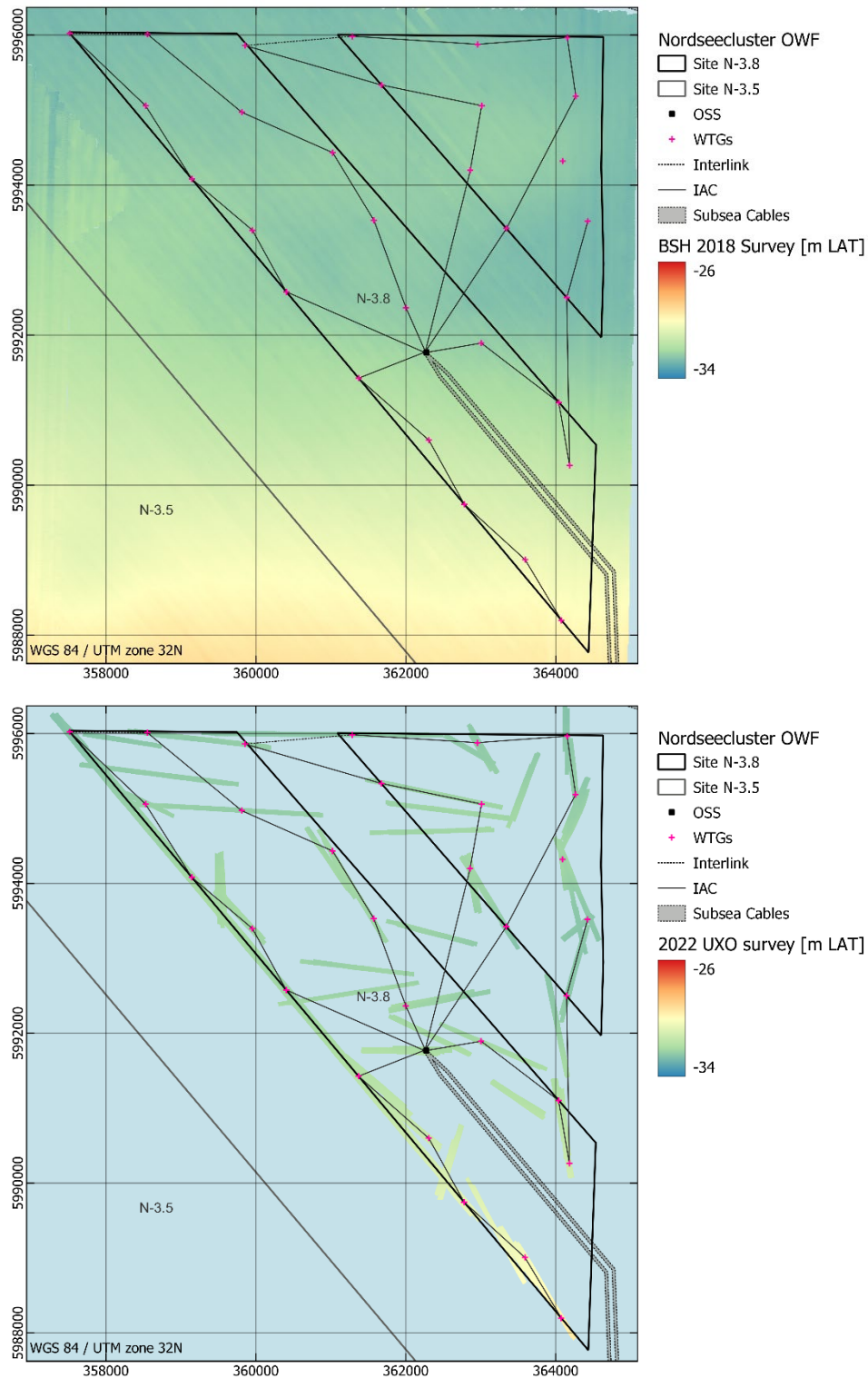


Figure 2.1 Top panel: Overview of the BSH 2018 bathymetry over Site N-3.8. Bottom panel: Overview of the UXO 2022 bathymetry over Site N-3.8. The outline of Site N-3.8 is marked with a black line.

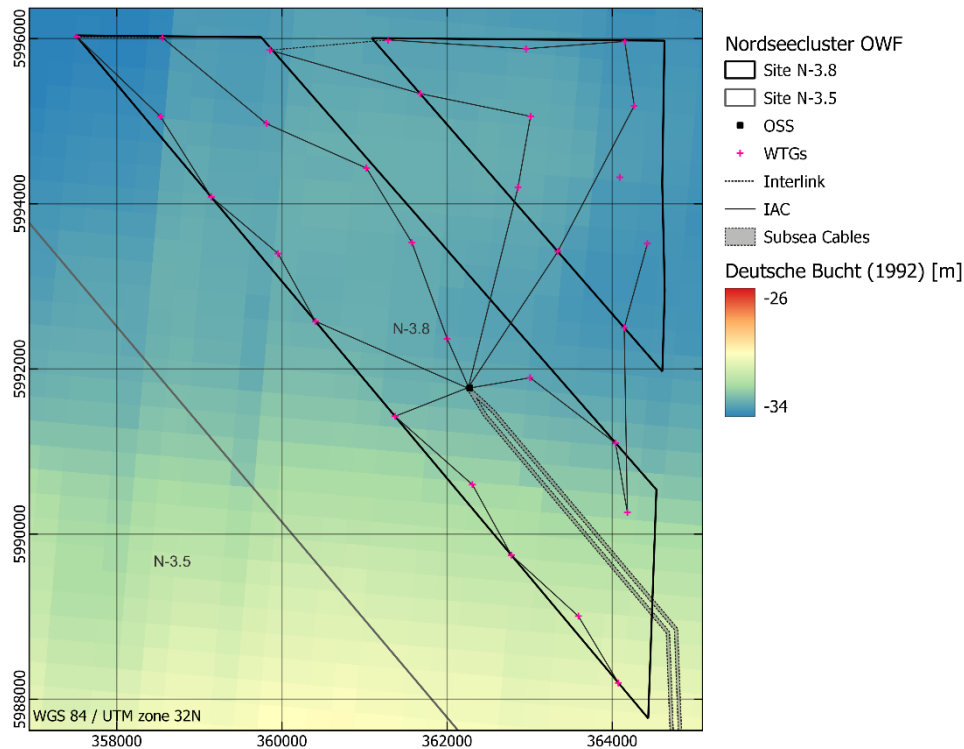


Figure 2.2 Overview of the 1992 Deutsche Bucht 3 dataset, around Site N-3.8, interpolated over a 300 x 300 m grid. The outline of Site N-3.8 is marked with a black line.

2.3 Geotechnical and geophysical data

Geological information was derived from Ramboll (2022) and the accompanying data. The data includes a geological model of this area, derived from hydrographic, geophysical and geotechnical surveys carried out from 2018 to 2019 on behalf of the BSH as well as other available data (e.g. geotechnical surveys carried out between 2007 and 2012). The geological model and accompanying report describe seafloor conditions, stratigraphic layer sequences, geomorphological structures and anomalies present at Site N-3.8.

Additionally, the geotechnical data available consists of:

- i. borehole and cone penetration tests at 27 locations in Site N-3.8 conducted on behalf of BSH in the periods 2007-2012 and 2018-2019 with target depth of ranging from 45 to 62 m (Bundesanstalt für Wasserbau (BAW), 2021);
- ii. cone penetration tests at 38 locations including the proposed WTG locations, conducted by Gardline in 2022 (Gardline, 2022a, 2022b); and
- iii. the 2016 relatively coarse petrographic and sedimentology maps within the German Exclusive Economic Zone (EEZ) in the North Sea (Sievers, 2020).

The boreholes allow classification of the sediments at different depths (lithology, basis sedimentary structures). The CPT logs provide geotechnical characteristics of the sediments to the penetration depth (e.g., tip resistance and sleeve friction values) allowing for the interpretation of the vertical sediment profile properties, mainly lithology and stiffness. The sedimentology map of the German EEZ is used to get an impression of the seabed for the wider area around the Nordseecluster OWF.

Figure 2.3 shows the locations of CPTs and boreholes located in and around Site N-3.8. This figure also contains an overview of the grain sizes for the wider area around the OWF.

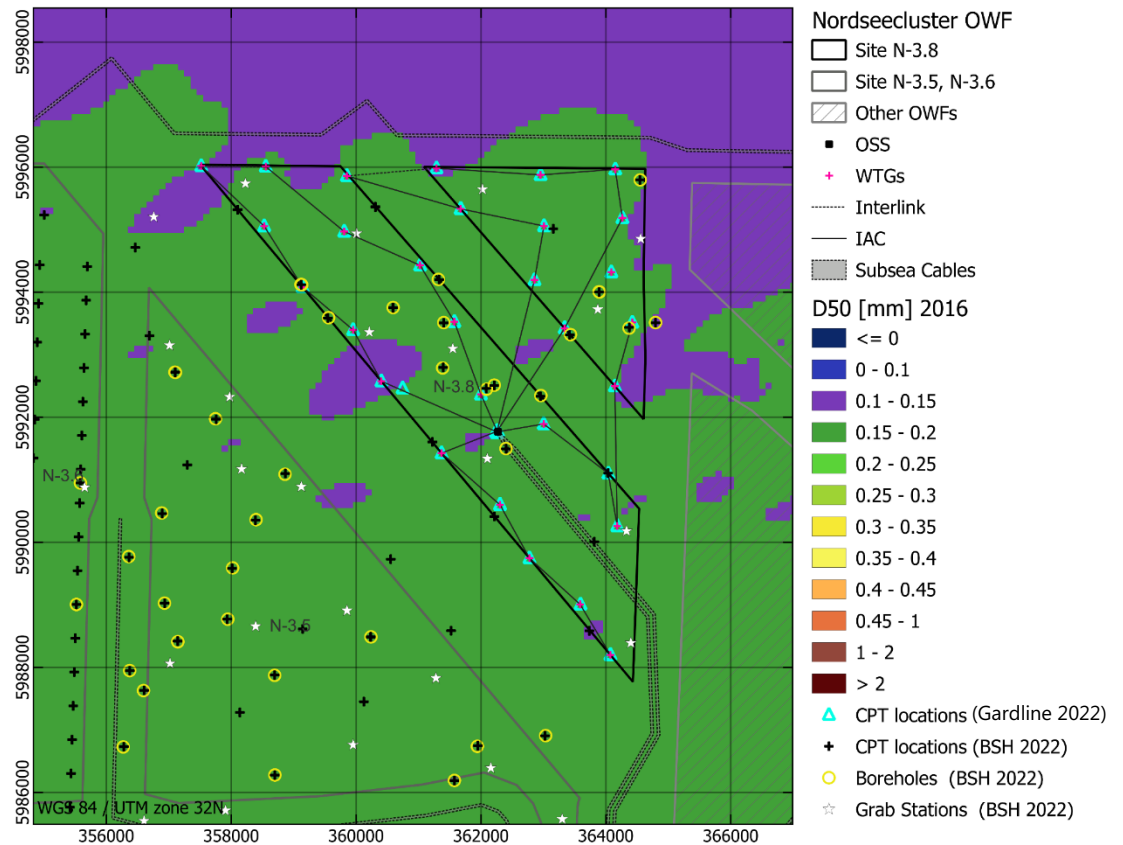


Figure 2.3 Borehole, Grab sample, CPT locations and an overview of the median grain size on the seabed in and around Site N-3.8 (Bundesanstalt für Wasserbau (BAW), 2021, Gardline, 2022a, 2022b, Sievers, 2020).

2.4 Hydrodynamic and wave data

The main sources of hydrodynamic and wave data for this study are the metocean reports for Site N-3.8 (ABPmer, 2023c, 2023d, Bundesamt für Seeschifffahrt und Hydrographie (BSH), 2021). The ABPmer (2023c, 2023d) metocean design reports contain detailed analyses of the wind, hydrodynamic and metocean conditions for three locations (Figure 2.4) based on long-term hindcast numerical modelling validated using local hydrodynamic and wave measurements.

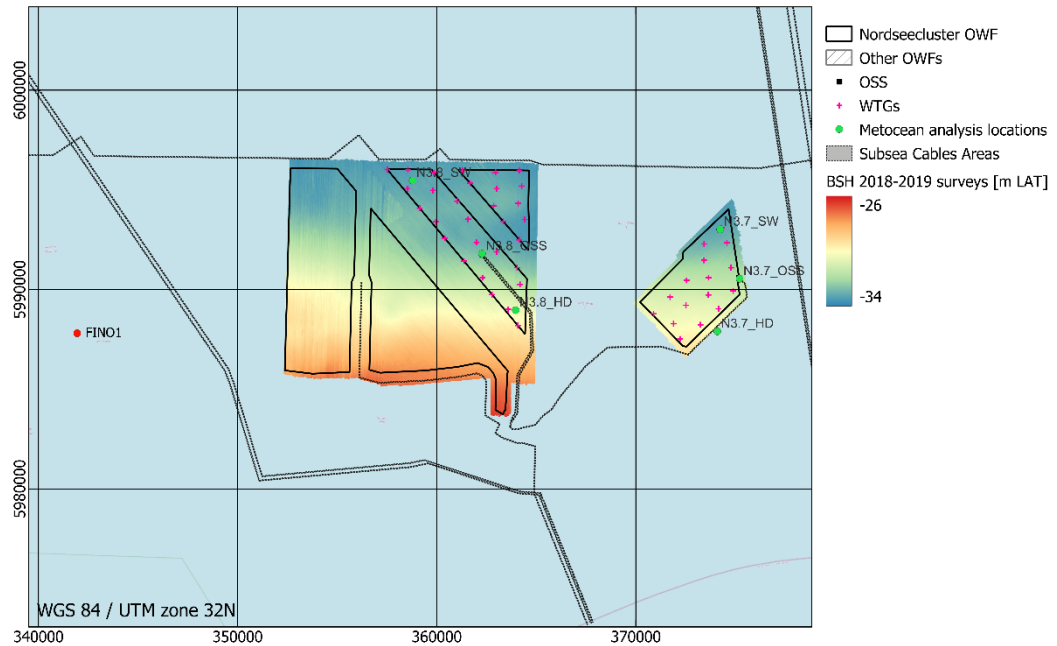


Figure 2.4 Overview of the metocean analysis locations for Sites N-3.8 and N-3.7 in (ABPmer, 2023a, 2023b, 2023c, 2023d). The labels SW and HD indicate wind and hydrodynamic analysis locations respectively. Both wind and hydrodynamic analyses were conducted for the OSS locations. Lastly, the location of FINO1 platform is marked with a red circle.

Figure 2.5 shows the roses of the surface (top of the water column) and depth-average current velocity and direction for normal conditions at the locations FINO1 and area N-3.8 (N3.8_HD in Figure 2.4) respectively. Similarly, the roses for the significant wave height and direction are shown in Figure 2.6.

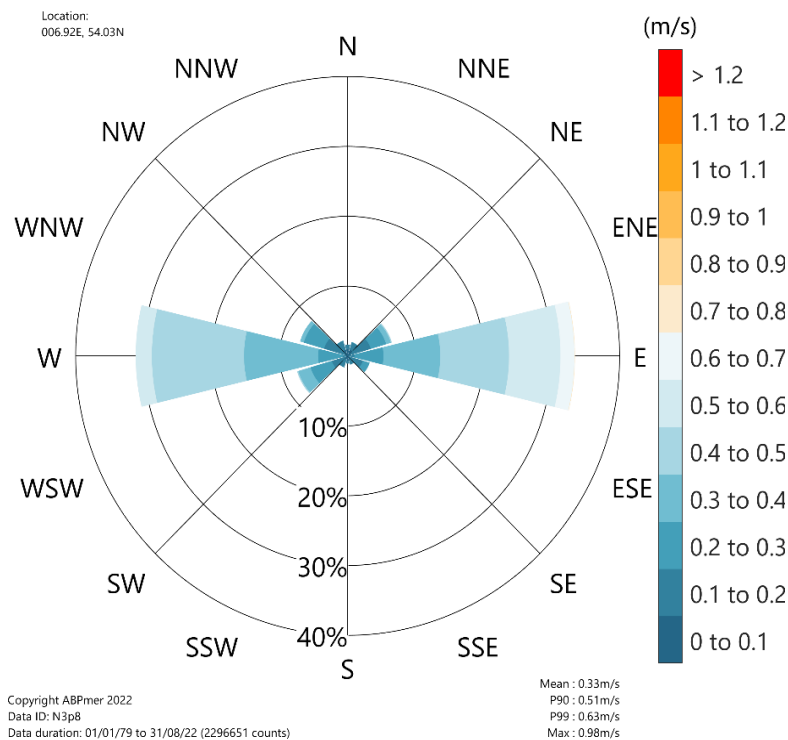
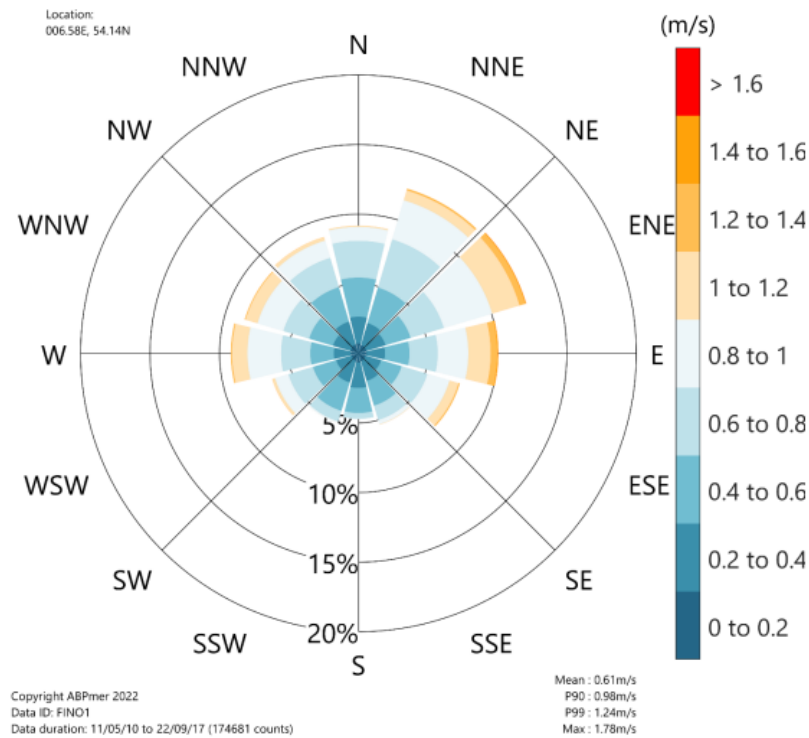


Figure 2.5 Top panel: Annual current velocity at surface (top of the water column) at FINO1. Bottom panel: Annual total depth averaged current velocity at point N3.8_HD (ABPmer, 2023d). Current directions correspond to the directions the currents are moving towards.

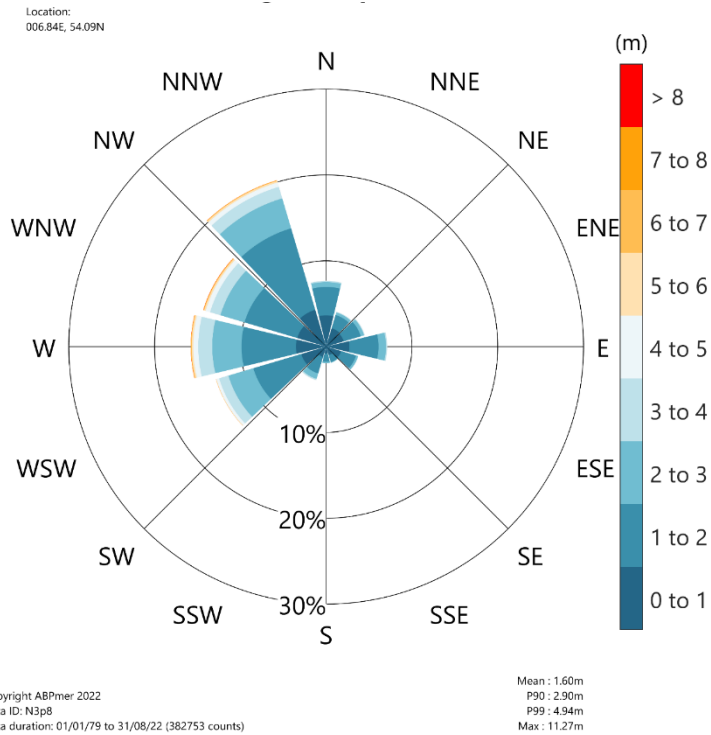


Figure 2.6 Annual significant wave height rose at location N3.8-SW (ABPmer, 2023d). Wave directions correspond to the direction the waves are coming from.

2.5 Other data

Bathymetric datasets of two neighbouring areas are considered in the present study:

- Five bathymetric datasets from the years 2017 to 2021 are available for the neighbouring windfarm site Nordsee One. The windfarm has been fully commissioned since 2017 around the time of the first dataset acquisition. The surveys cover the inter-array cable corridors and WTG locations with an average width of 150 m. All surveys are of high resolution – 0.5 x 0.5 m.
- One full covering survey from the years 2019 is available for the neighbouring Site N-3.7. Lastly, a 2022 survey of specific locations over Site N-3.7 is available.

The above datasets are used to investigate the seabed characteristics and any bed level changes at these neighbouring locations and support the seabed level predictions for Site N-3.8. For more details on the available surveys and seabed mobility analysis of Site N-3.7, reference is made to Deltares (2023).

3 Seabed mobility study

3.1 Introduction

The objectives of the seabed mobility study are to assess the morphodynamics and provide an estimation of future seabed levels around Site N-3.8. To achieve these objectives, an analysis is performed in the following parts: an overview of the hydrodynamic, wave, sediment and sediment transport characteristics in and around Site N-3.8 is given in Section 3.2; the morphodynamic analysis based on the system understanding and data analysis is presented in Section 3.3. The final sections, 3.4, 3.5 and 3.6, include the future bed level estimates, the UXO analysis and risk assessment respectively.

3.2 Overview of the characteristics in and around Site N-3.8

Site N-3.8 of Nordseecluster OWF is located in the North Sea, approximately 55 km north off the German coast in the German Bight. The area is characterised by tide- and meteo-driven west to east sediment transport, a sloped seabed with minor undulations and seabed substrate that predominantly consists of fine to very fine sand. This section gives an overview of the hydrodynamic, wave, sediment transport and the geotechnical and geophysical characteristics in Site N-3.8 and the wider German Bight region.

3.2.1 Hydrodynamic and wave characteristics

A combination of several processes is driving the sediment dynamics in Site N-3.8 of Nordseecluster OWF. These governing processes include hydrodynamics (such as tides and wind-driven flow) and waves. Some of these processes are driven by atmospheric variations and meteorology (such as wind patterns, pressure fields, precipitation), river discharge (from Weser, Elbe and other rivers) which impose seasonal patterns as well as episodic extreme events on the system.

The German Bight is a meso-tidal to macro-tidal environment, with a tidal range between 1.5 m to approximately 3.5 m (at the Elbe river mouth). The semi diurnal tides enter the German Bight from the west and propagate counter-clockwise along the coast towards Danish waters (Zeiler et al., 2014, Zeiler et al., 2008). Figure 2.5 shows the roses of the depth-averaged total and surface total current velocity and direction for normal conditions at one location within Site N-3.8 (N3.8_HD in Figure 2.4) and FINO1 respectively. The roses in this figure show that the depth-averaged currents are generally below 0.5 m/s, with a mean magnitude of 0.33 m/s and predominantly oriented towards the east or west direction. Larger directional variability and larger amplitudes are observed for the surface currents at FINO1, under the effect of wind. The monthly roses (included in ABPmer (2023c, 2023d)) show limited seasonal variability in the depth-averaged velocities in N-3.8. The surface currents (top of the water column) at FINO1 vary monthly between west to north to east directions under the effect of varying wind conditions. The strongest currents (>1.6 m/s) are observed during January and are ENE directed.

Figure 2.6 shows the significant wave height and direction for one location within Site N-3.8. The significant wave heights are generally below 2.90 m (for 90% of the time). The associated peak period is below 8 seconds. The monthly wave roses in ABPmer (2023c, 2023d) show that for the majority of the year waves approach from the W – NNW, with the highest waves observed during winter (December to February).

3.2.2 Sediment transport

The fine surface sediment of the German Bight can be mobilised under the effect of waves and residual currents. Diesing et al. (2006) estimated that fine sand sediments at locations with 30 m depth off the North Frisian coast can become mobile under the effect of waves and tide for approximately one quarter of the year.

Figure 3.1 presents the average suspended and bed load sediment transport magnitude and direction in the German Bight simulated under the effect of tide, surge, river run-off, wind, and waves over one year as part of the AufMod research program (Heyer & Schrottke, 2013; Heyer et al., 2014). As can be seen in the figure, along the East Frisian coast, north of the barrier islands, sediment transports are generally west to east directed. Further east within the German Bight, sediment transports have varying directions. At the deeper part of German Bight bed load sediment transports become very small, while suspended sediment transports gradually turn north-northwest.

Zeiler et al. (2014) described that within the German Bight the largest amount of sediment is transported along the foreshore, between 10 and 20 m of depth, while meteorological forcing (wind & waves) has a significant effect on the residual currents and sediment transport. Sediment transport pathways close to the shore are influenced by variations in river discharge and local bathymetry, while further offshore the sediment transport pathway are fairly stable over the over the years (Milbradt, 2012).

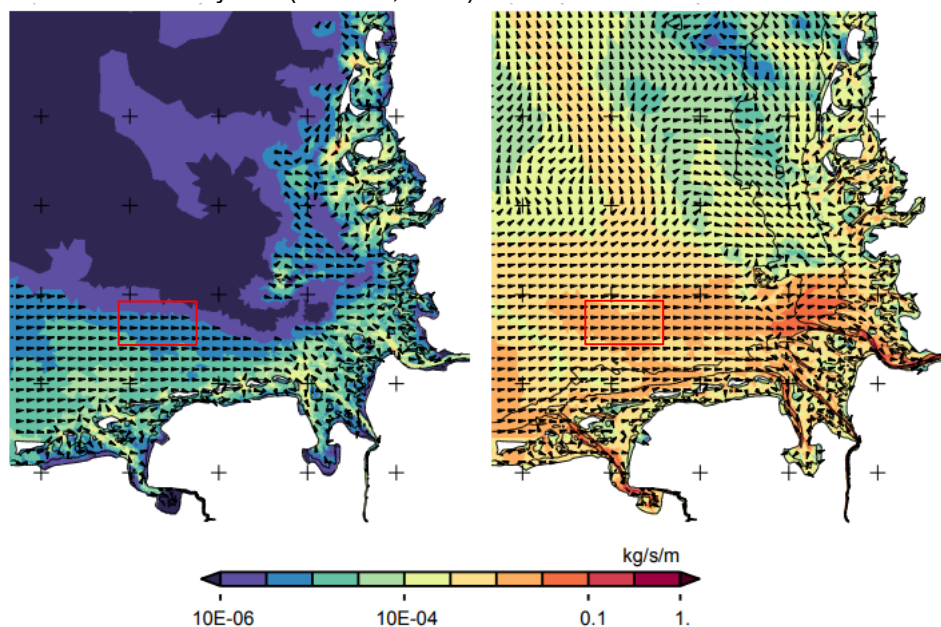


Figure 3.1 Residual bed load (left panel) and suspended (right panel) transport patterns (Heyer & Schrottke, 2013; Heyer et al., 2014). The approximate location of Nordseecluster OWF is marked on the map with the red polygons

Under the dominant hydrodynamic conditions, the German Bight is a sediment accumulation area. Figure 3.2 presents an overview of the in literature available estimates of sediment transports in the North Sea (Zeiler et al., 2014). In the German Bight, sediment is imported primarily from adjacent North Sea regions. Additionally, riverine discharge of fine sediments contributes significantly to the bed sediment in the German Bight (Plüß, 2009). Furthermore, Zeiler et al. (2008) note that during storms, sediment from the shallow shoreface can be transported offshore up to 40 m of depth. Reineck (1963) and McCave (1970) estimate deposition rates between 20 cm and 27 cm per 100 years over the German Bight respectively.

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m depth off the North Frisian coast can become mobile under the effect of waves and tide for approximately one quarter of the year.

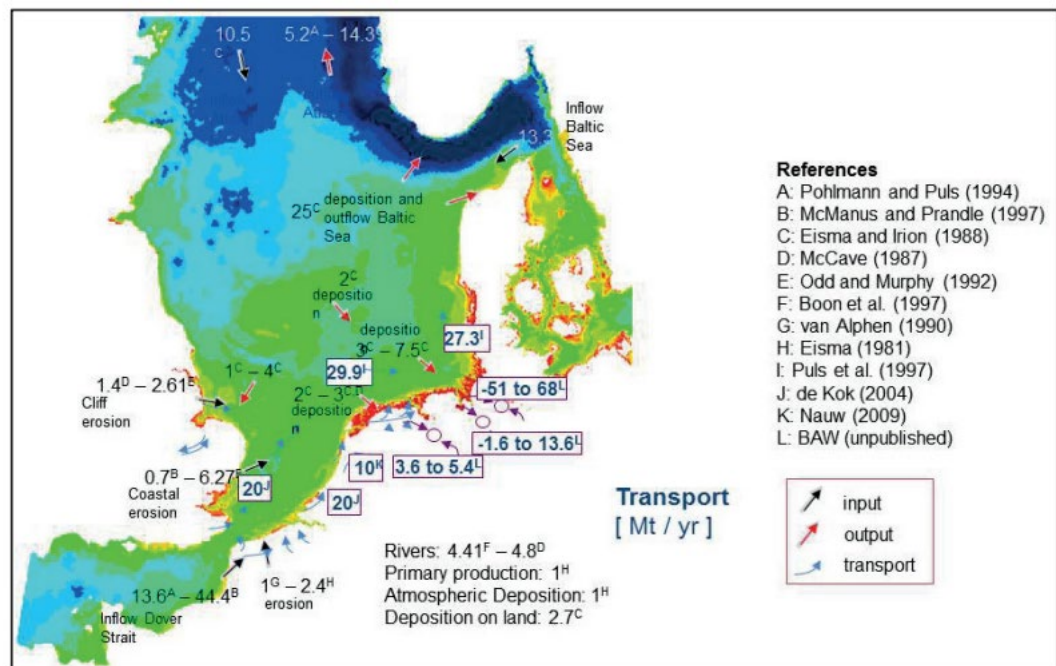


Figure 3.2 Estimates of sediment transport budgets in literature (Zeiler et al., 2014).

3.2.3 Geotechnical and geophysical characteristics

3.2.3.1 Regional scale

According to Ramboll (2021) the geology of the southern German Bight is primarily composed of Holocene and Pleistocene sediments, which were deposited over the last 2 million years. The fluctuations in sea levels during the Pleistocene glacial and inter-glacial periods caused the formation of various types of deposits in the area including glaciogenic, marine and terrigenous. The rise of sea level during the Holocene period led to the creation of a shallow sea in the German Bight and deposition of fine-grained sediments. These sediments include sand, silt, and clay, and were formed by a combination of processes such as river input, wind transport, and reworking by ocean currents and waves.

Figure 3.3 presents the distribution of median sediment grain sizes on the seafloor over the southeast part of the German Bight (Sievers, 2020). The majority of the area is covered by sand (with median grain sizes (d_{50}) 0.063 to 2 mm). Gravel ($d_{50} > 2\text{ mm}$) is found in only a few areas and consists of relict Pleistocene deposits and is considered stable in their position under the hydrodynamic regime (Zeiler et al., 2000). Close to the Nordseecluster OWF the sandy deposits consist mainly of fine sand (d_{50} of 0.063 to 0.2 mm) with occasional patches of medium sand.

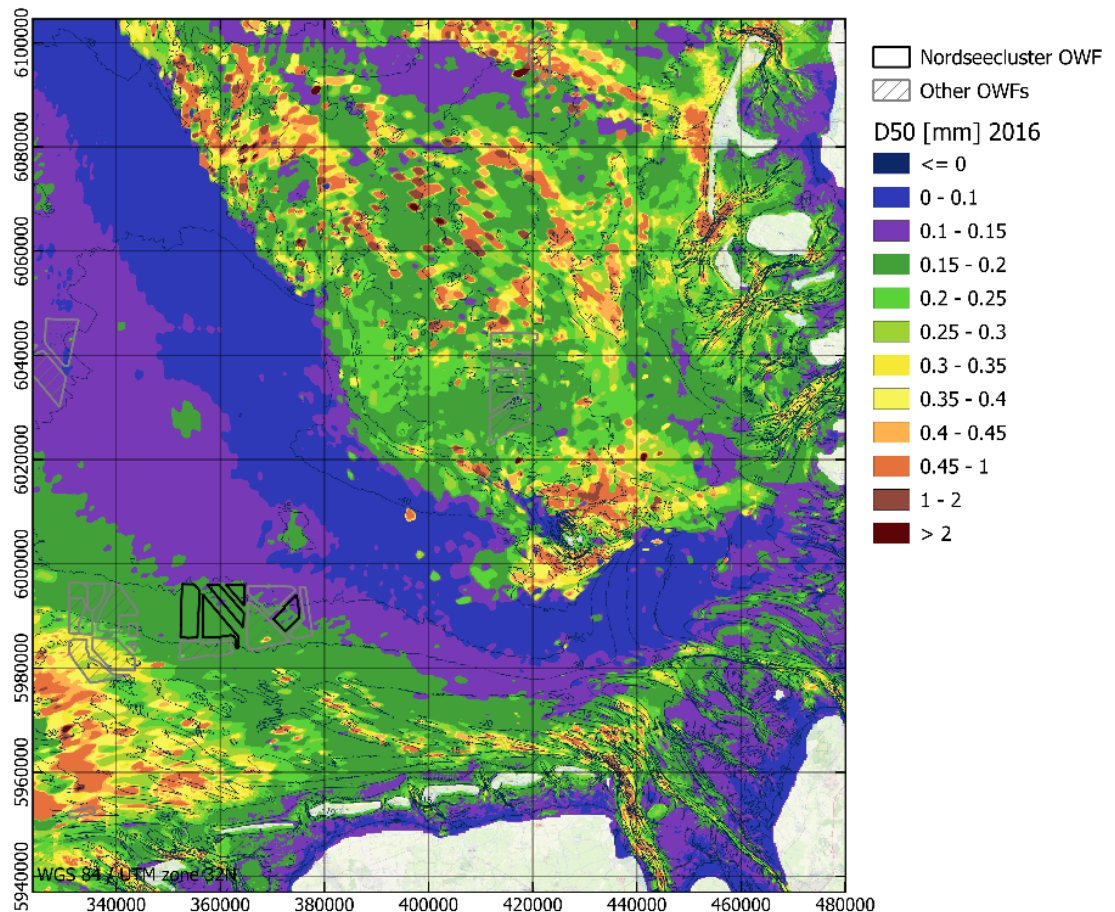


Figure 3.3 Overview of the median grain size distribution over the German Bight (Sievers, 2020). Nordseecluster OWF sites are marked with black polygons. Other windfarm sites are marked with grey polygons.

3.2.3.2 Nordseecluster OWF -Site 3.8

Six distinct geological units have been identified in Site N-3.8 according to Ramboll (2022):

- The top unit (Unit Ia) consists of Holocene marine deposits of very loose to medium dense sand, mud and peat. For the majority of Site N-3.8 the thickness of the unit ranges between 0 and 2 m (Figure 3.4, top panel). The thickness of the soil unit increases locally up to 11 m at the locations of the two large-scale depressions and some point depressions. A meandering depression approximately 5m deep, occurs at the southeast part of Site N-3.8 and a southwest to northeast depression, approximately 7m deep, crosses Site N-3.8 at approximately the middle latitudes;
- Unit Ib is found underlying Unit Ia. It consists of dense to very dense fine to medium-coarse Pleistocene sands with sporadic instances of thin layers of peat. The thickness of the unit ranges between 0 and 23 m;
- Unit II consists of interglacial silt and clay and dense-very dense glacial sand originating in the Pleistocene age. Over the general area of Site N-3.8 the thickness of Unit II ranges between 7 and 27 m, with larger thickness (up to 122 m) observed within a large-scale valley depression. The depression, with maximum depth of 100 m and southwest to northeast direction crosses Site N-3.8 at approximately the middle latitudes and digs into the underlying soil Units. Within this depression a thick layer (0 to 122 m) of river infill -soft silt and clay occurs and is described as Unit IIa (Figure 3.4,

bottom panel). Outside of the depression, Unit II is mainly sandy consisting of fine to medium grained sand.

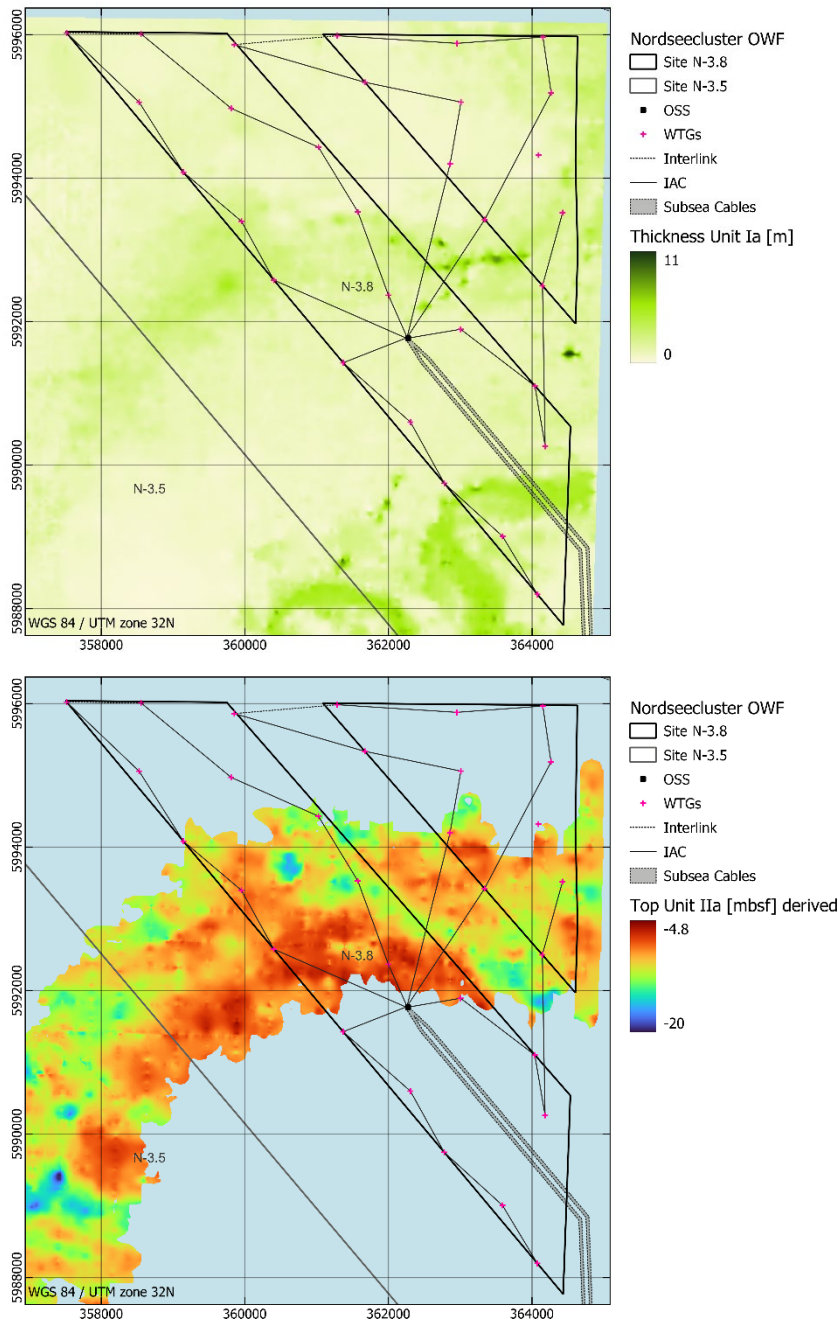


Figure 3.4 Top panel: Thickness of geological Unit I (Ramboll, 2022) over site N-3.8. Bottom panel: Derived top of the Unit IIa (Ramboll, 2022) over site N-3.8.

- Unit III underlies Unit II with a thickness varying between 0 m (at the location of the valley depression described above) and 59 m. It consists of medium grained very dense Pleistocene sands with occasional occurrence of gravel and thin peat layers;
- Unit IV is described as a mixture of sand, clay, silt organic matter and peat. It is observed locally at a few borehole locations with a thickness of maximum 4 m;
- Lastly, Unit V underlies Units II, III and IV. It consists of medium dense fine to coarse grained sands with occasional incidence of gravel and thin layers of organic matter, clay and peat.

Along with the hydrodynamics, the type of sediment present on the seabed and its properties are a determining factor for sediment transport and seabed morphodynamics. An overview of the median grain diameter along with the geotechnical measurement locations in and around Site N-3.8 is shown in Figure 2.3. This figure shows that the sediment in Site N-3.8 is predominantly classified as fine sand with decreasing median diameter towards the northeast. The borehole, grab sampling and CPT logs of Bundesanstalt für Wasserbau (BAW) (2021) and Gardline (2022a, 2022b) comply with these observations. These show the top layer consists of sand mixtures of fine and medium sand. Occasionally layers of peat or low strength clay are observed close to the surface (within 4 metres below the seabed). Deeper, layers of medium to coarse sand are found. Thicker layers of clay or silt are found deeper than 6 metres below the seabed for some of the survey locations.

3.2.4 Seabed characteristics

This section discusses the seabed characteristics within the German Bight, in Site N-3.8 and within Nordsee One OWF.

3.2.4.1 Regional analysis

The bathymetry of the south eastern German Bight is shown in Figure 3.5. Water depths range between 0 m at the coast and 70 m (south of the Helgoland islands). Close to the coast (until approximately the 15 m waterdepth contour line) the bathymetry is influenced by the presence of a series of islands, ebb tidal deltas and tidal channels. At the southwestern part of the presented bathymetry, south of Nordseecluster OWF, the bathymetry becomes gradually smoother with increasing water depths. The presence of elongated sand ridges/ribbons is evident at the southwest directed 'tongues' of the 20 and 25 m contour lines. From the 30 m contour line the seabed becomes smoother and north-northeastwards sloping.

In Figure 3.6 a hillshaded composite bathymetry of the area around Nordseecluster OWF is presented. This visualisation emphasizes slopes, making it easier to identify features such as bedforms present on the seabed. Close to Nordseecluster OWF two areas with bedforms are observed and marked with yellow circles. Southwest of the OWF area, shallower than -27 m MSL (approximately -25.5 m LAT) southwest to northeast directed bed forms are observed. The bed forms have heights in the order of 1 m and lengths between 200 and 1000 m. When considered in combination with Figure 3.3, it is observed that there is a correlation with the presence of the bedforms southwest of the OWF area and the larger grain sizes present in the seabed. Additionally, bedforms are observed west of Nordseecluster OWF at depths similar to those in the OWF. The bedforms are southwest to northeast directed with heights in the order of 1 m and lengths between 700 and 1100 m. With the exception of these two areas, no indication of bedforms is found in or around the Nordseecluster sites. It is noted though that the resolution of the bathymetry composite is too limited to get indications of the presence of smaller bedforms such as megaripples.

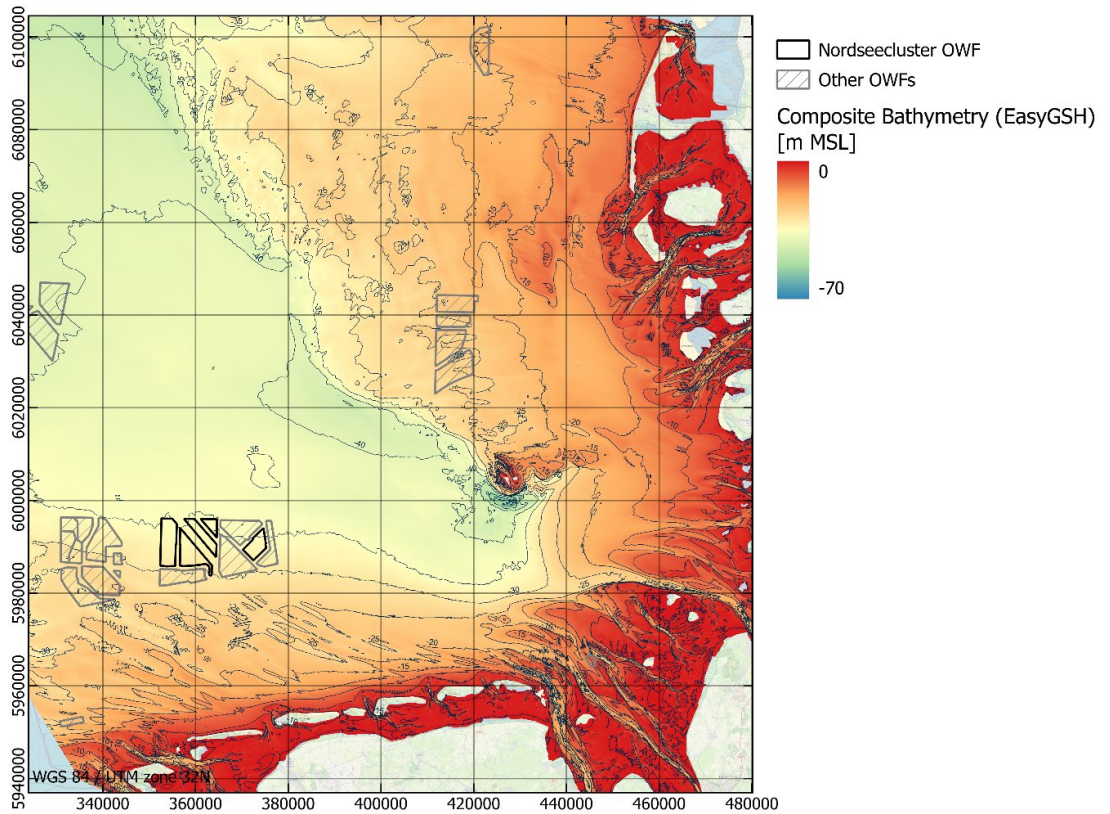


Figure 3.5 Overview of the bathymetry over the German Bight. The location of Nordseecluster OWF has been marked with black polygons. The location of other windfarms is marked with grey polygons.

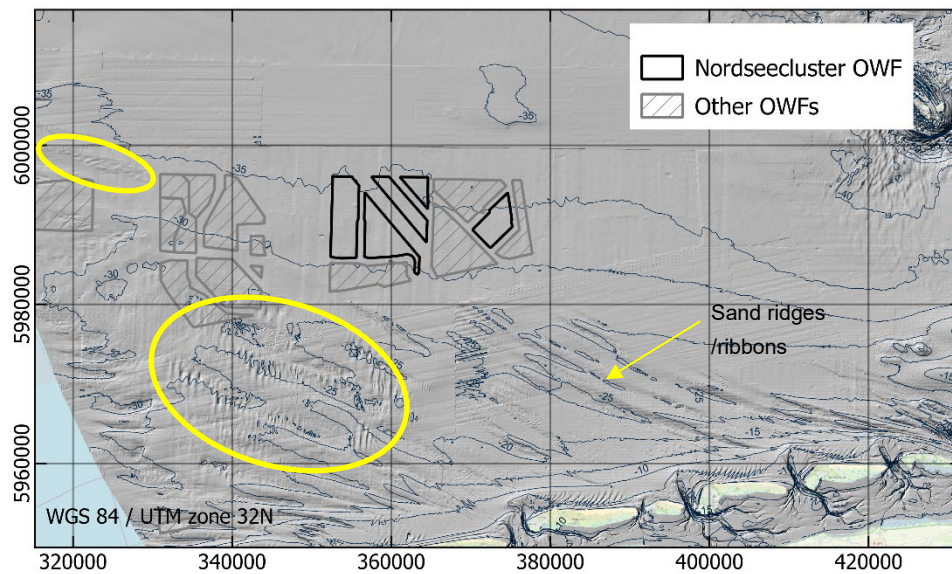


Figure 3.6 Hillshaded view of the bathymetry at the southwest part of the German Bight including waterdepth contour lines. The location of Nordseecluster OWF has been marked with black polygons. The location of other windfarms is marked with grey polygons. Yellow circles indicate areas with bedforms.

3.2.4.2 Nordseecluster OWF -Site 3.8

The seabed in Site N-3.8 can be characterised as a mildly sloping seabed, with an overall dip from south to north, covering a total elevation span of approximately 6 m. Locally local minor undulations are observed. An overview of the bathymetry is presented in Figure 2.1 and Figure 2.2. Figure 3.7 presents a closeup of the hillshaded BSH 2018 bathymetry, overlaid with the

MBES track lines (red lines). The survey presents repeating bed level undulations, contained between consecutive track lines and with (along-crest directions perpendicular to the direction of the track lines).

For example, in the top panel of Figure 3.7 these undulations are observed mainly in the northeast to southwest direction (along-crest direction), perpendicular to the southeast to northwest tracklines' direction. However, at the south west part of the presented area (outside of Site N-3.8) these undulations appear in the west to east direction, perpendicular to the north to south now directed track lines. These undulations typically have heights in the order of a few centimetres and length in the order of tens of metres (see also Figure 3.10 transects A and B). Due to their characteristics and orientation, they are considered as survey artifacts potentially due to errors in the pre-processing/calibration or variations in the sound velocity during the MBES survey campaign, similar to the artifacts described in Hellequin et al. (2003) and Bian Gang (2018).

In Figure 3.7 the lower panel gives an overview of the bed level variation over the smooth sloping seabed extracted from the BSH 2018 survey, using an averaging filter with a circular base of 1000 m. This variation includes the heights of both types of survey artifacts described above as well as any pockmarks. For the majority of the Site, this variation is within the ± 0.1 m range.

The survey gives no indication of the presence of bed forms such as sand waves and megaripples. It is noted though, that the presence of small-scale bed forms could be concealed by the survey artifacts.

The UXO 2022 survey although with limited spatial coverage, shows a smooth sloped seabed. Figure 3.8 shows parts of the hillshaded 2022 survey. The survey shows some small-scale undulations, with directions that are not always perpendicular to the directions of the surveyed lines, suggesting that those undulations are bedforms rather than survey artifacts. The height of these bedforms (see also Figure 3.10) and any other bed level roughness of the seabed is within ± 0.05 m.

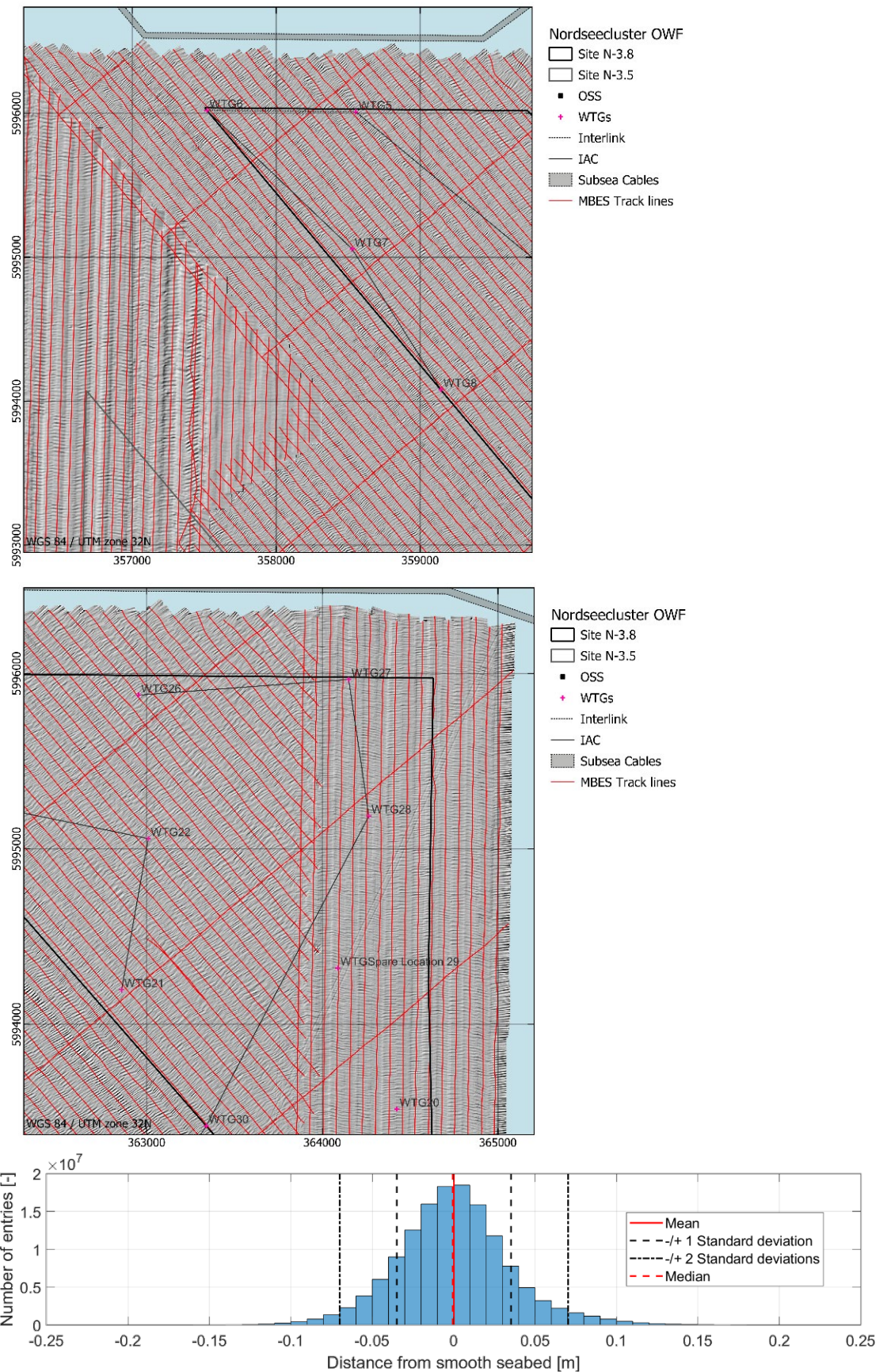


Figure 3.7 Top and middle panel: Hillshaded views of the 2018 BSH bathymetry. The MBES tracklines are marked as red lines. Bottom panel: Distribution of bed forms' undulations' seabed roughness heights in the 2

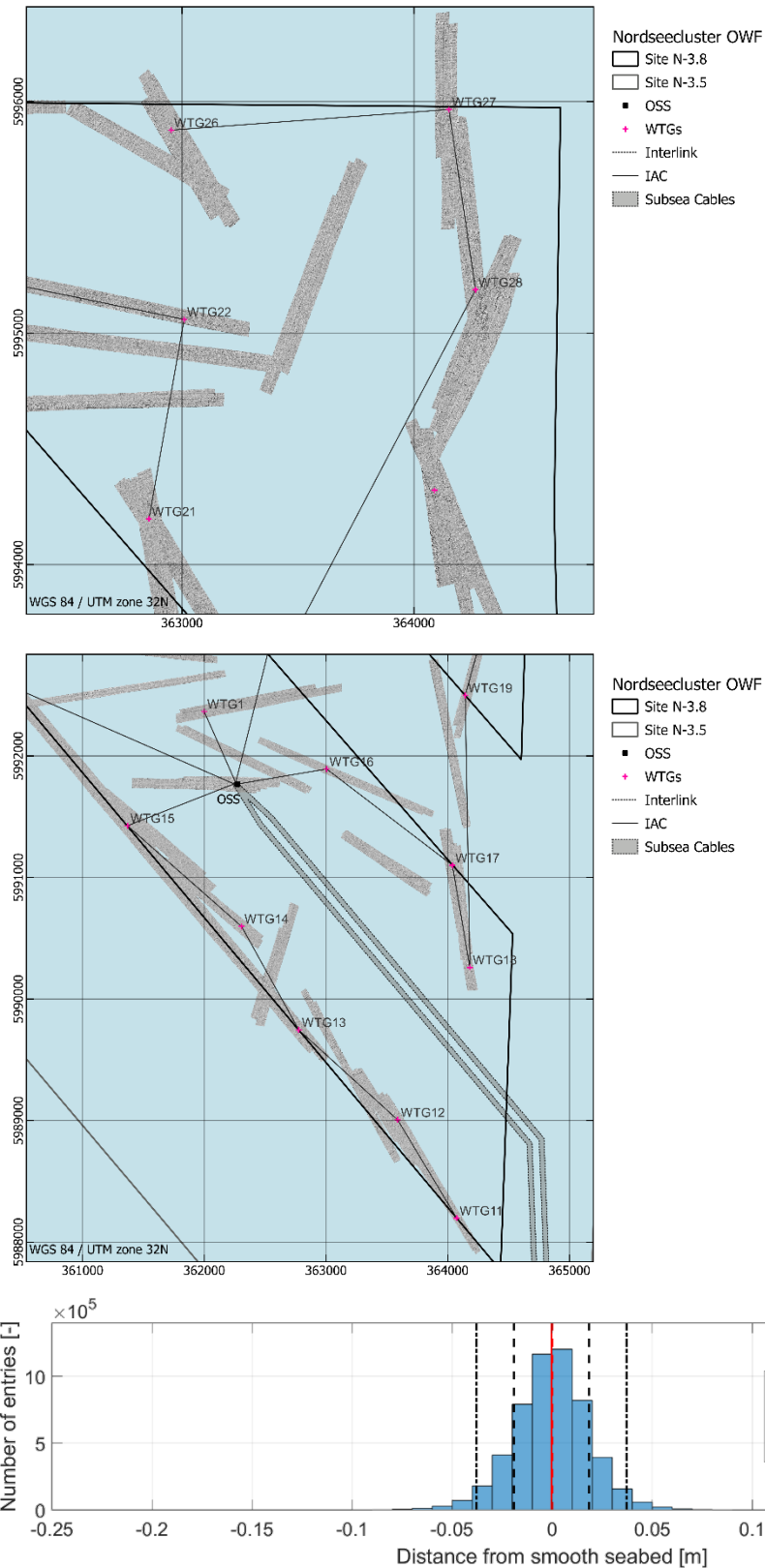


Figure 3.8 Top and middle panel: Hillshaded views of the 2022 UXO bathymetry. Bottom panel: Distribution of bed forms/ undulations/ seabed roughness heights in the 2022 UXO bathymetry.

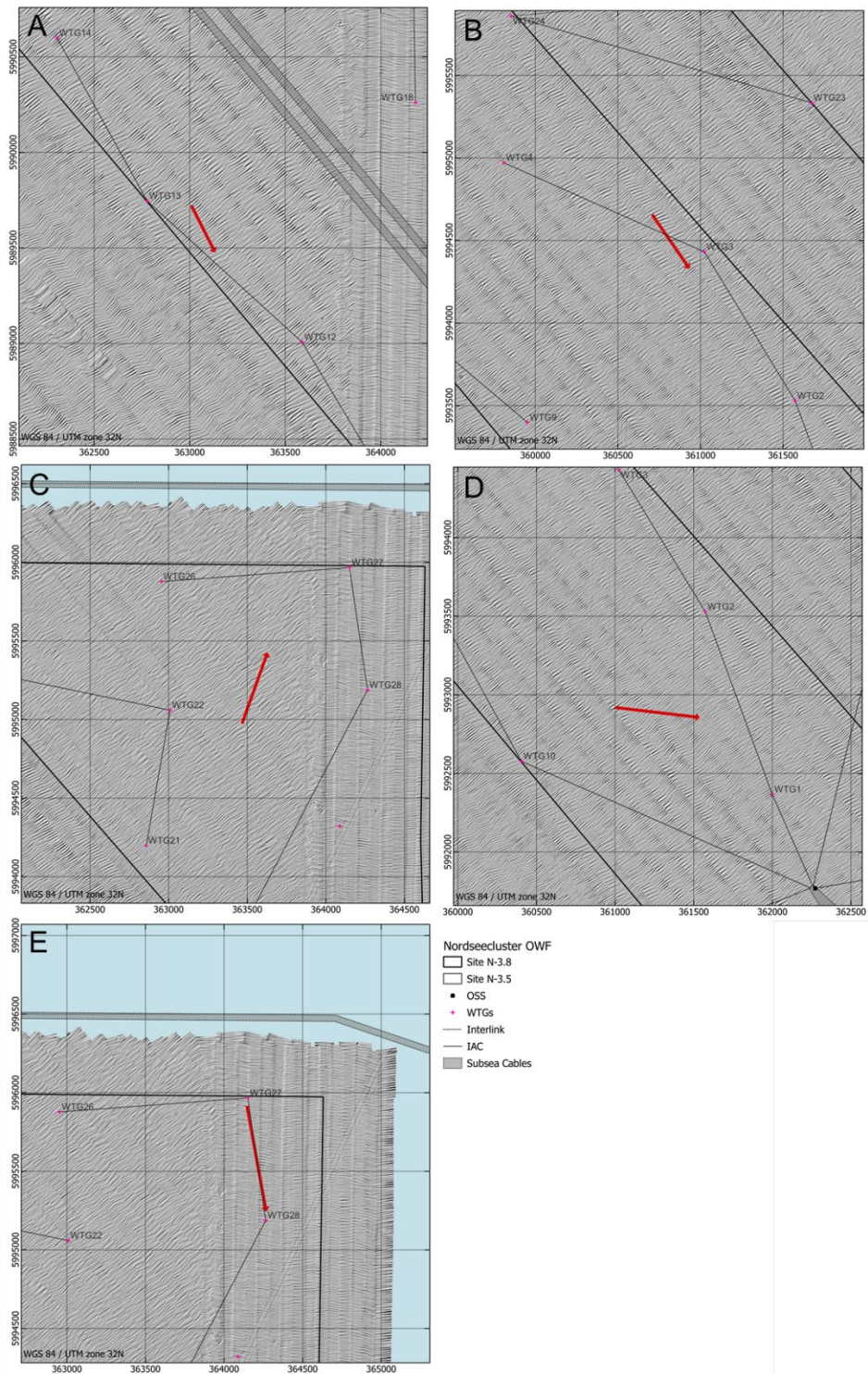


Figure 3.9 Location of four transects over Site N-3.8. On the background the 2018 BSH bathymetry is presented.

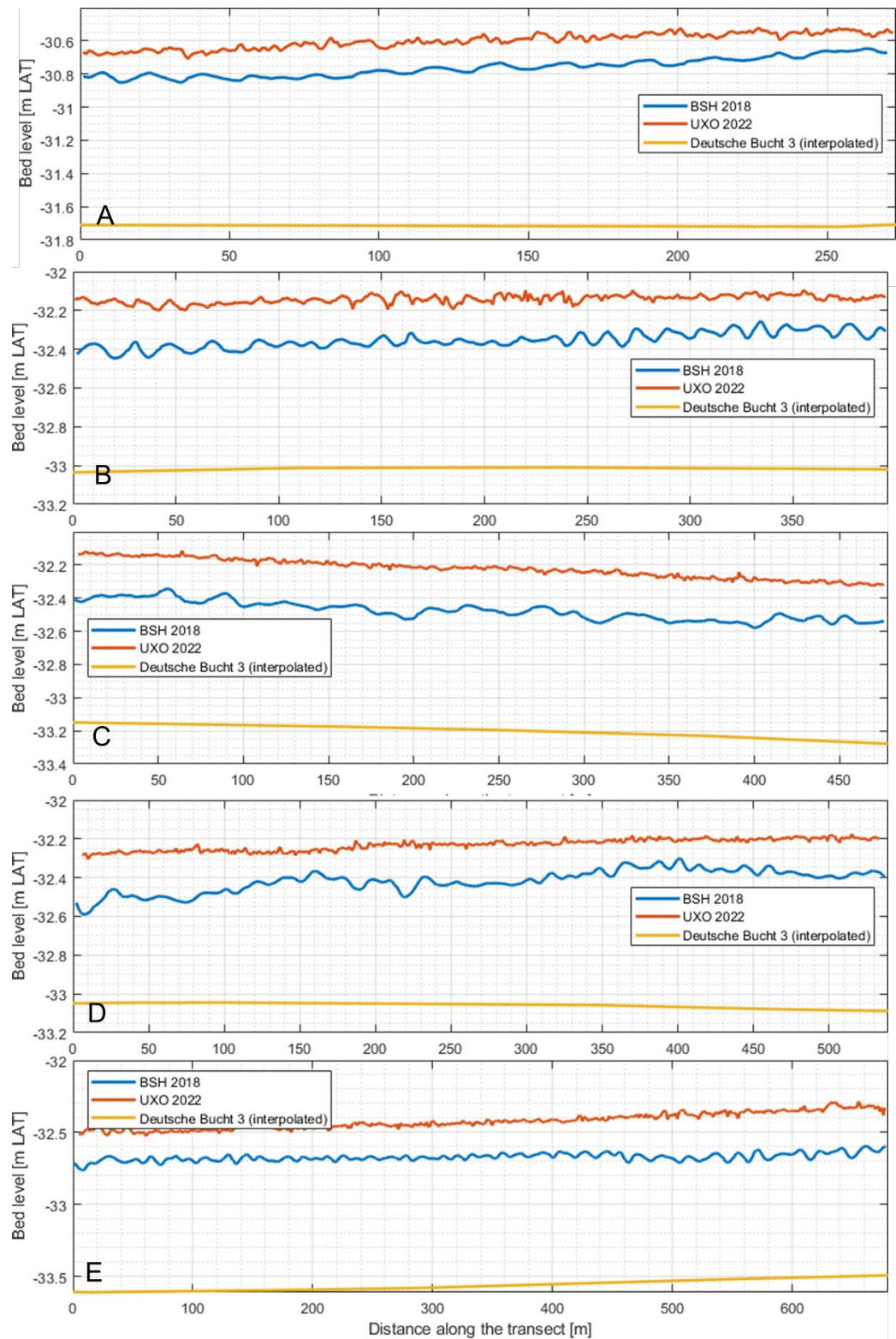


Figure 3.10 Measured bed levels along five transects in Site N-3.8. The location of the transects is presented in Figure 3.9. No corrections have been made on the surveys.

3.2.4.3 Neighbouring sites

The analysis of the seabed characteristics over the adjacent Nordsee One OWF is performed to gain further insight in seabed dynamics close to Site N-3.8. The bathymetry over the Nordsee One area is dominated by the presence of human interventions. As seen in Figure 3.11, close to the location of the WTGs, the shape of the scour protection as well as edge scour are visible.

Traces of the inter-array cable routes are observed over the 2017 bathymetry and for a few locations (such as the one marked in Figure 3.11) over all bathymetries.

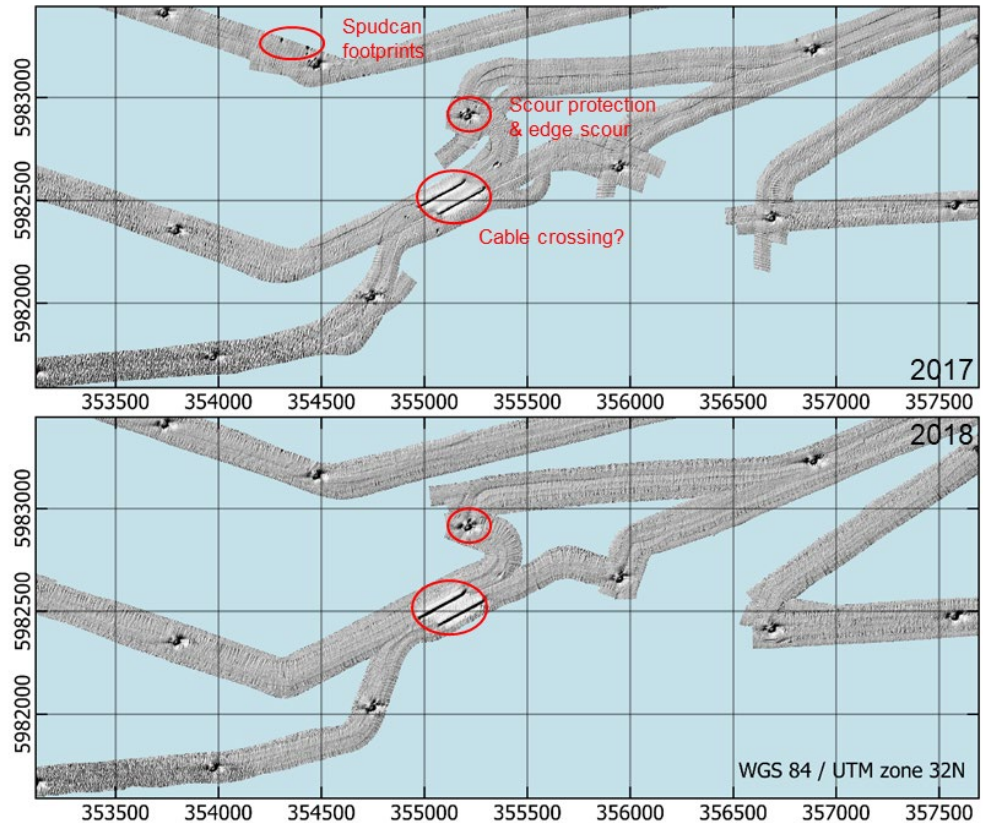


Figure 3.11 Hillshade view of the 2017 (bottom panel) and 2018 (top panel) surveys over part of the Nordsee One OWF. The locations of spudcan footprints, WTG scour protection and edge scour as well as a (possible) cable crossing are marked in the two bathymetries.

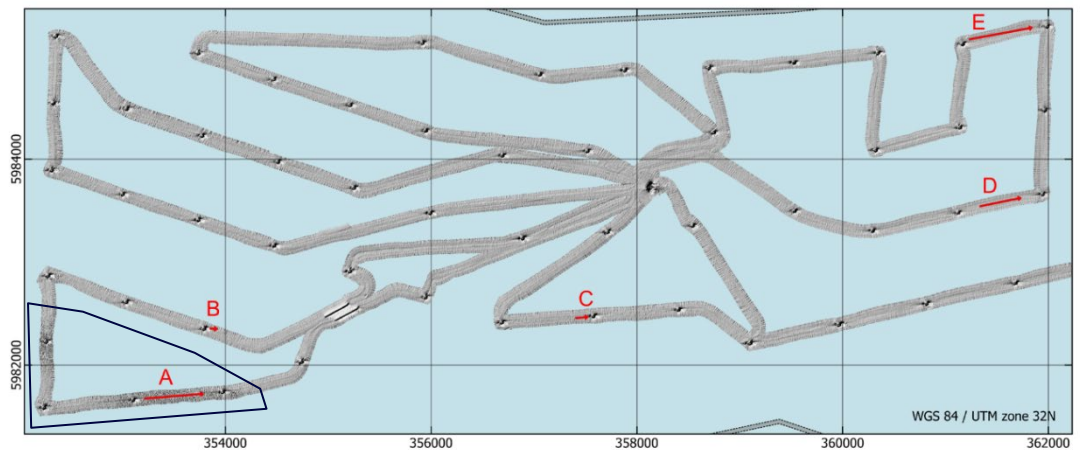


Figure 3.12 Hillshaded view of the 2021 bathymetry over Nordsee One OWF. The locations of four transects are marked with red arrows. The blue polygon marks the area where dense megaripples are observed on the seabed.

Additional to the human interventions, small bedforms are observed at the southwest edge of the survey area in Nordsee One (transect A in Figure 3.13). The bedforms observed have height up to 0.3 m and lengths in the order of tens of meters (20-40 m). The crest of these bed forms is north-south directed, perpendicular to the dominant current directions. These bed forms occur densely at the area marked with the blue polygon in Figure 3.12. Their dimensions

vary slightly in the different surveys. Outside of these areas, similar dimension bedforms are sometimes observed at the east (downstream) side of the WTGs, at the edge of the edge scour holes (transect B in Figure 3.13). At fewer locations bedforms are observed at the west side of the WTG locations (transect C in Figure 3.13). Towards the north and east sides of Nordsee One, the seabed becomes gradually smoother and any bedforms present decrease in size (transects D & E in Figure 3.13). Some survey artifacts can be observed at the edges of the surveyed corridors. The survey artifacts, any small-scale bed forms observed in the area (usually close to the WTGs) and bed form roughness are within ± 0.05 m of bed level variations.

The analysis of the available datasets over N-3.7 in Deltares (2023) shows a relatively flat seabed, with no indications of the presence of bedforms such as sand waves or megaripples.

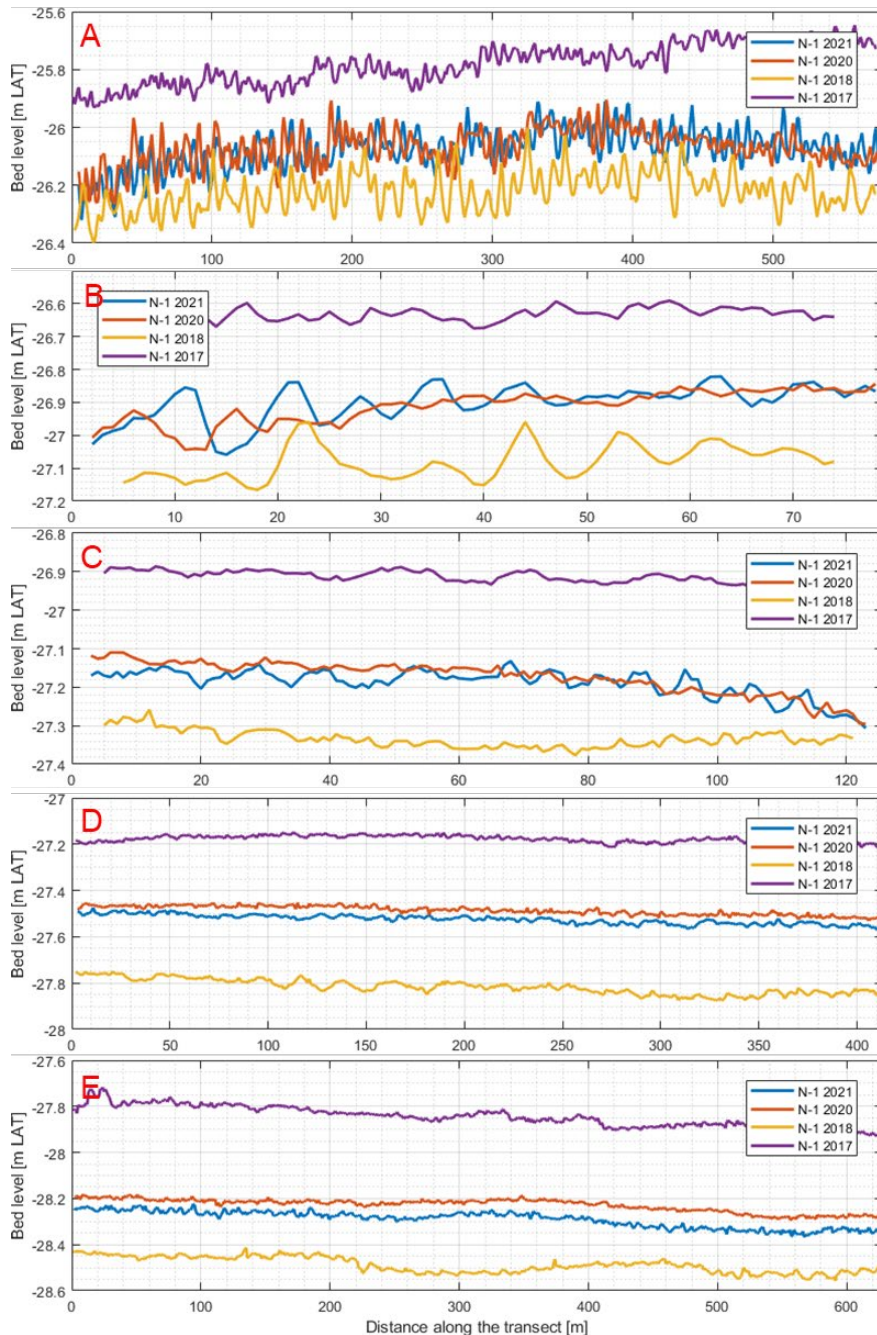


Figure 3.13 Measured bed levels along five transects in Nordsee One OWF. The location of the transects is marked in Figure 3.12. No corrections on vertical levels have been applied on the surveys.

3.3 Morphodynamic analysis

This section presents the morphodynamic analysis in and around Site N-3.8. First a data-driven analysis is presented focussing on seabed changes within Site N-3.8. The analysis is supplemented with estimates of seabed changes on a regional-scale available in literature, as well as with the insights from data analysis on neighbouring Sites/ OWFs.

3.3.1.1 Seabed level changes in Site N-3.8

To create an understanding of seabed level changes in and around Nordseecluster Site N-3.8 comparisons of the available bathymetric data are made. This analysis aims to determine if bathymetry changes have occurred over time and derive trends of bed level change that may apply in the future.

Figure 3.14, and Figure 3.15 present an overview of differences in seabed levels at Site N-3.8 based on the comparison of the available bathymetry datasets presented in Table 2.1. All comparisons are made with respect to the highest resolution, full covering bathymetry (i.e. the BSH 2018 bathymetry). No corrections have been applied on the datasets during this first step. Additionally, for the comparison of the BSH 2018 bathymetry with the Deutsche Bucht 3 (1992) dataset, the pre-interpolation (raw) version of the latter dataset is used, so as to avoid introducing errors through the interpolation process. The differences that can be observed in these figures fall within the vertical uncertainties of measurements (International Hydrographic Organization, 2020). They might not be related to seabed dynamics but might just be the result of uncertainties and differences in the vertical reference levels of the surveys.

Figure 3.16 contains two histograms that show the distribution of seabed level differences for the bathymetry comparisons presented in Figure 3.14, and Figure 3.15. This figure also shows the mean, median and standard deviation for the seabed level differences.

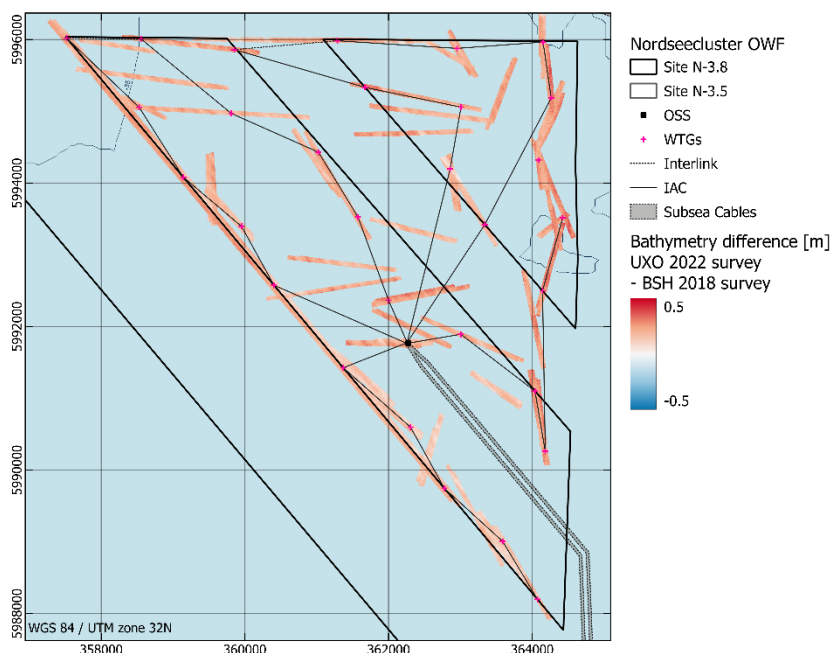


Figure 3.14 Difference between the UXO 2022 and the BSH 2018 survey over Site N-3.8.

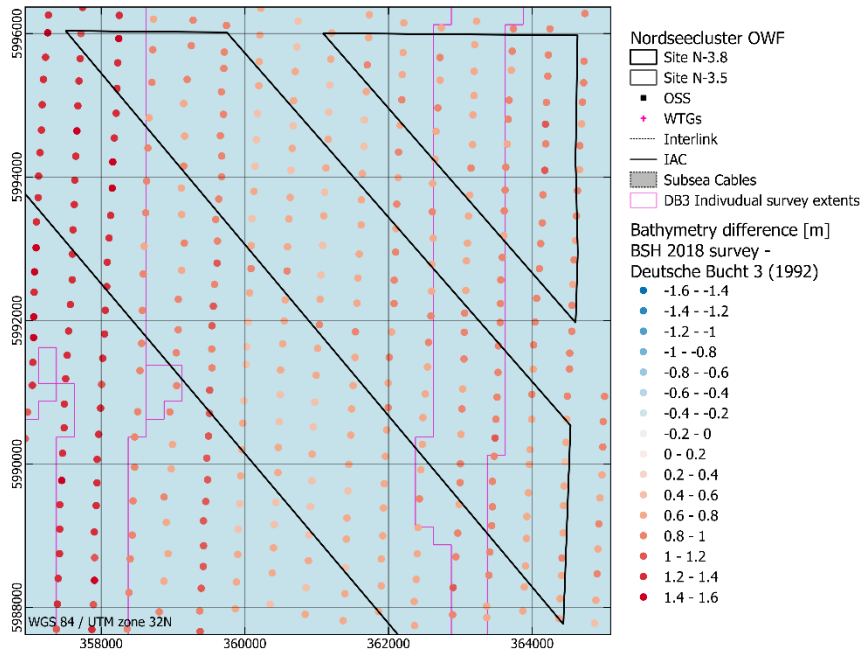


Figure 3.15 Difference between the BSH 2018 and the Deutsche Bucht 3 1992 dataset over Site N-3.8. The magenta lines indicate the extents of the different surveys that make up the Deutsche Bucht 3 dataset.

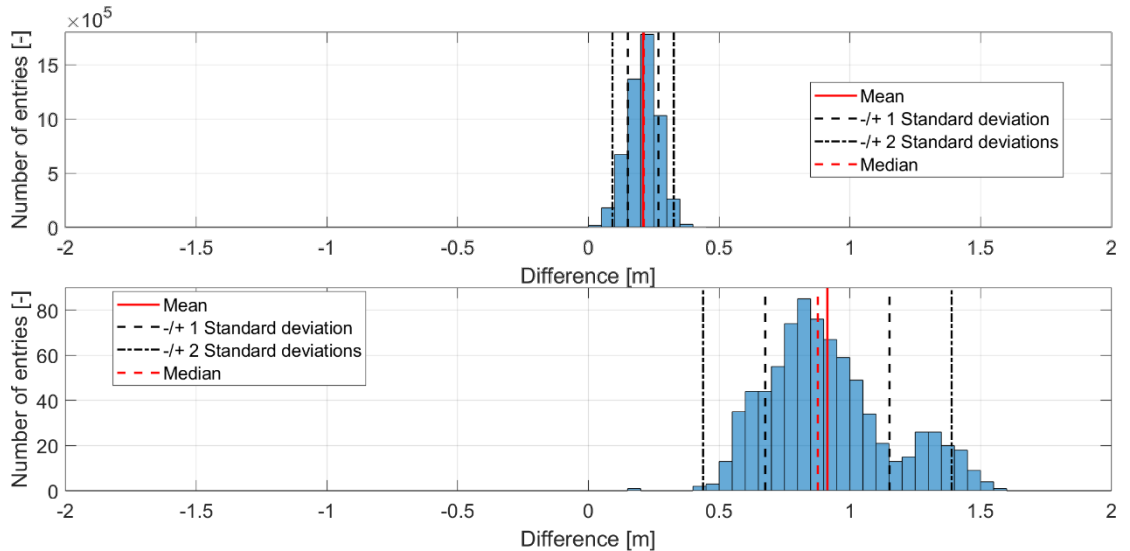


Figure 3.16 Distribution of the bed level differences between the 2022 UXO survey and the 2018 BSH survey (top panel) and between the BSH 2018 survey and the 1992 Deutsche Bucht 3 dataset (bottom panel)

From the overview figures (Figure 3.14 and Figure 3.15), the histograms (Figure 3.16) and the bed levels over transects (Figure 3.10), it can be observed that there are relatively constant offsets between the BSH 2018 survey and each of the other two bathymetry datasets. These offsets are in the order of a few decimetres, with 0.9 m and 0.21 m on average, when comparing to the Deutsche Bucht 3 (1992) and the UXO 2022 datasets respectively. The bed level differences are generally normally distributed around the mean with standard deviation of 0.23 and 0.06 m respectively. These offsets, especially the offset from the 1992 dataset, are most likely caused by differences in survey equipment, data processing methods and referencing techniques used when creating the different datasets. However, the possibility that the observed differences relate to actual seabed level changes, although unlikely, cannot be ruled out from the analysis of the available data.

To get insight in bed level changes related to morphodynamic activities in the area, corrections are made to the datasets. The aim of these corrections is to discover whether apart from the uncertainties in bed level measurements and differences in vertical reference levels any other local bed level change has occurred in the area. A spatially uniform correction, equal to the mean of the differences (see Figure 3.16) is applied to the oldest and newest bathymetries. After correction the mean of the differences for each bathymetry comparison is equal to zero. This means that the Deutsche Bucht 3 (1992) dataset is increased by 0.9 m and the UXO 2022 survey is increased by 0.21 m.

The overviews of the differences in seabed levels in Site N-3.8 after correction are presented in Figure 3.17 and Figure 3.18. When compared to the 2022 UXO bathymetry the 2018 BSH bathymetry was smoothed to remove from the bed level differences the spatial variation related to the presence of survey artifacts within the 2018 survey. A circular filter with a base diameter of 1000 m was used for this purpose. The histograms of the corrected differences are shown in Figure 3.19. The histograms for the difference between the 2018 and corrected 1992 bathymetries show that the majority of the observed differences (within two standard deviations) fall between -0.46 and 0.46 m. The differences between the 2022 and 2018 bathymetries are less widespread and generally fall (within two standard deviations) between -0.12 and 0.12 m.

The observed differences are very small, within the range of uncertainty of the survey data. Especially for the comparison between the two oldest datasets (Figure 3.18), the spatial variation of bed level changes correlates with the extents of the different datasets making up the 1992 Deutsche Bucht 3 dataset, suggesting that the remaining bed level differences are related to survey errors/offsets within the Deutsche Bucht 3 dataset, rather than morphodynamic activity. The comparison of the 2018 and corrected 2022 survey (Figure 3.17) shows seabed level lowering mainly at the southeast side of Site N-3.8 whilst seabed level rising is observed more prominently along the east side of Site N-3.8. The spatial distribution of these differences over the smooth, mildly sloping, seabed with relatively uniform seabed sediment and hydrodynamics resembles quasi-random survey errors and/or inaccuracy for the different survey patches, rather than patterns of morphodynamic activity. It is noted that the data available is not sufficient to fully confirm this hypothesis.

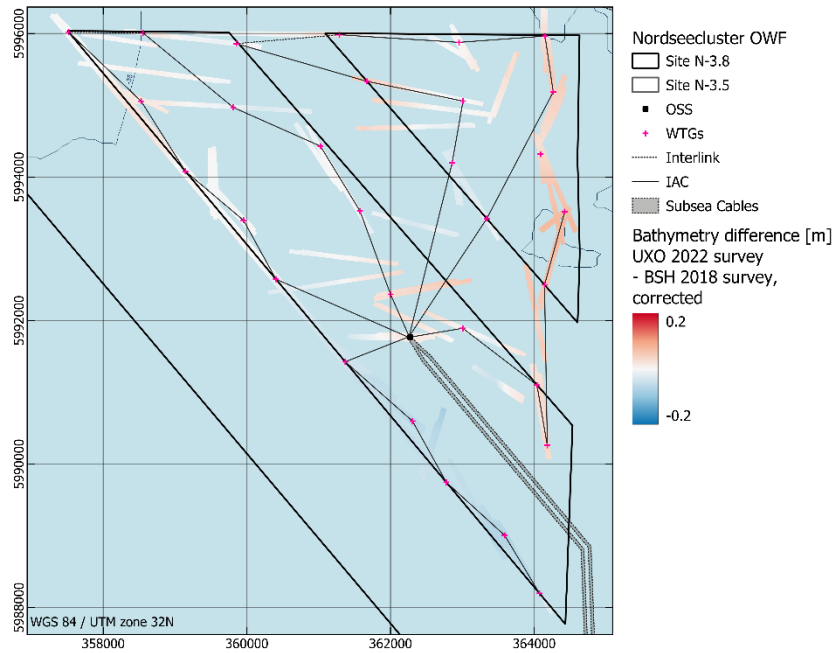


Figure 3.17 Difference between the corrected 2022 UXO bathymetry and the smoothed 2018 BSH bathymetry over Site N-3.8.

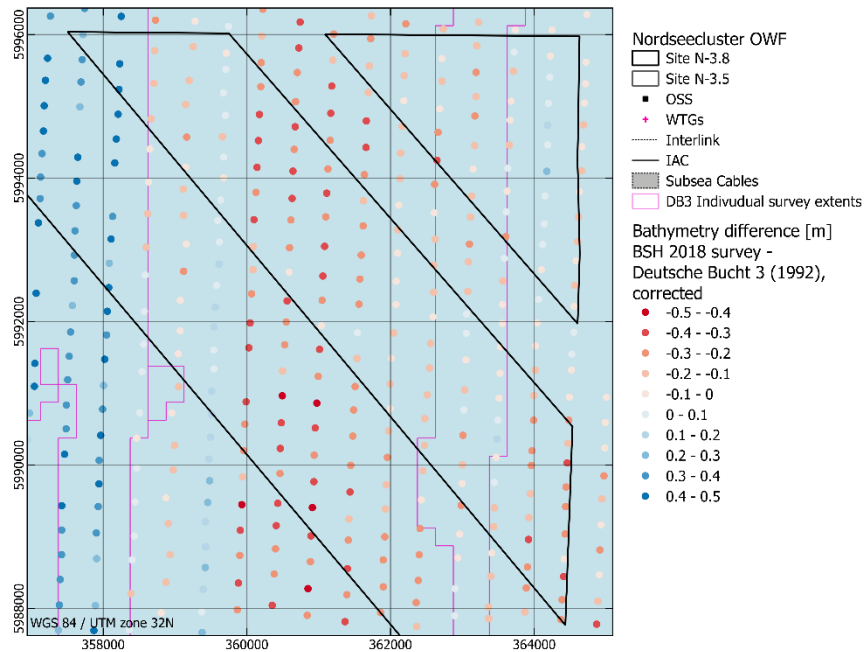
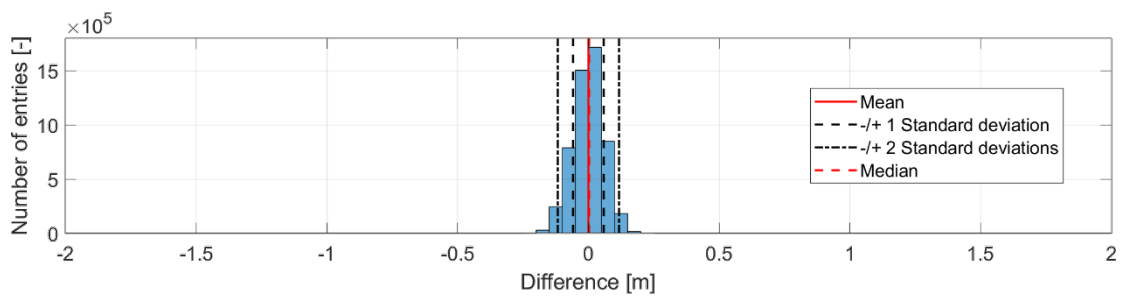


Figure 3.18 Difference between the 2018 BSH bathymetry and the corrected 1992 Deutsche Bucht 3 bathymetry over Site N-3.8. The magenta lines indicate the extents of the different surveys that make up the Deutsche Bucht 3 dataset.



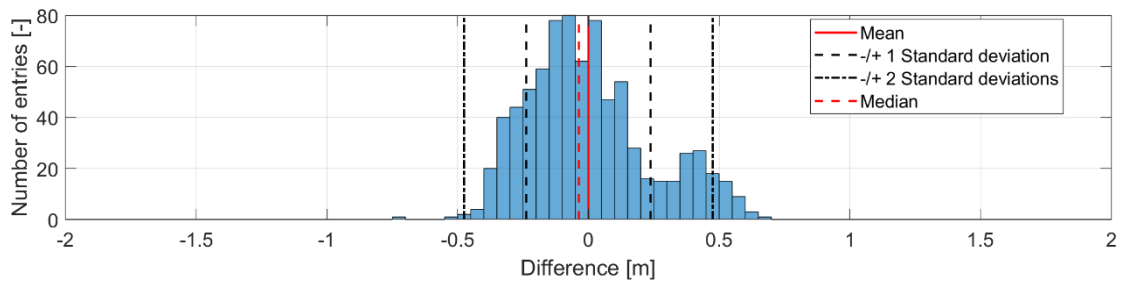


Figure 3.19 Distribution of bed level differences between the corrected 2022 UXO survey and the 2018 BSH survey (top panel) and between the BSH 2018 survey and the corrected 1992 Deutsche Bucht 3 dataset (bottom panel).

3.3.1.2 Regional analysis

As discussed in Section 2.3, the bathymetric data available outside and close to Nordseecluster OWF is limited to a single low-resolution bathymetric dataset from 1992 (Deutsche Bucht 3). However, for other areas in the German Bight more than one bathymetry dataset is available. For those areas, the difference between the maximum and minimum measured seabed level constitutes the morphological space (Bundesamt für Seeschifffahrt und Hydrographie (BSH), 2022). Figure 3.20 presents the morphological space for the southeast part of the German Bight as provided by BSH for a period of approximately 20-30 years (for the specific dates of survey reference is made to Appendix A). As can be seen from the figure, larger morphological change has been observed close to the coast, the barrier islands and tidal inlets. Moving seawards, the observed bed level changes reduce to a few decimetres. Areas that are completely white – such as at and around Nordseecluster OWF as well as the central and northwest part of the presented area – are areas for which only one survey dataset is available and thus the morphodynamic space cannot be derived.

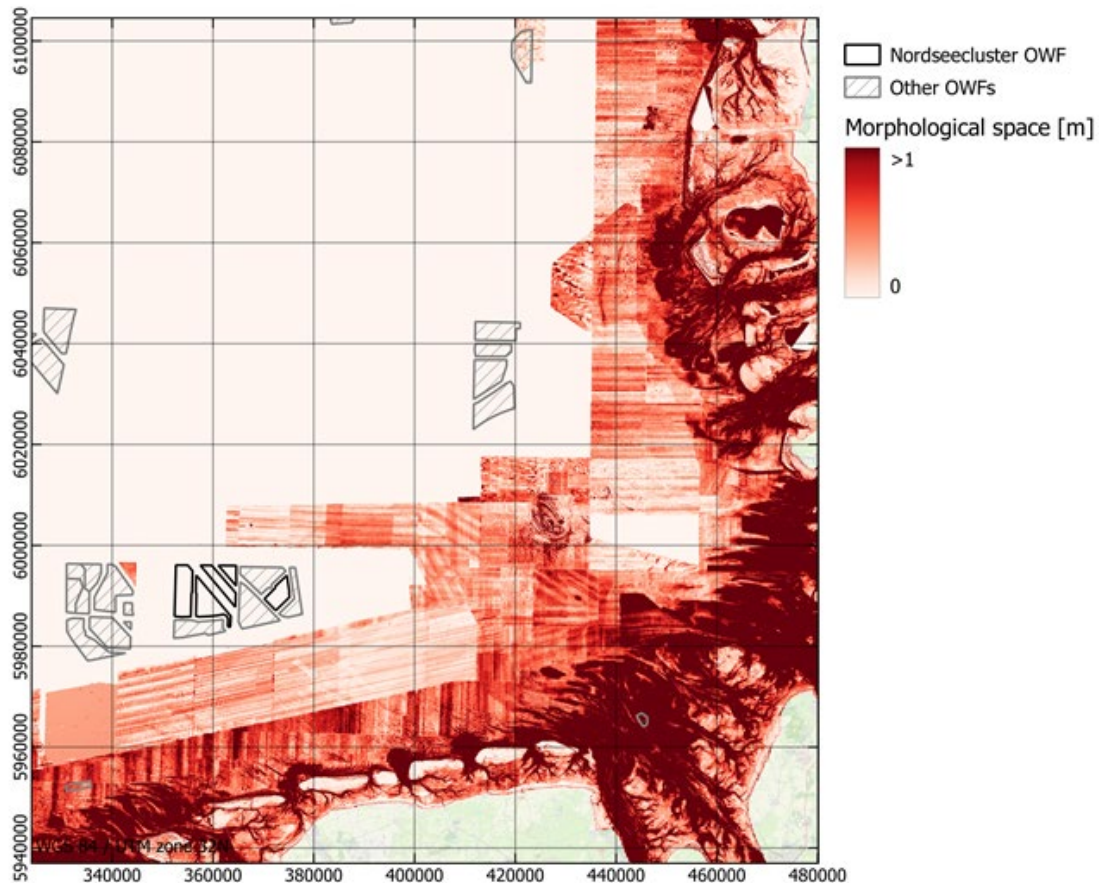


Figure 3.20 Overview of the morphological space over the German Bight (Bundesamt für Seeschifffahrt und Hydrographie (BSH), 2022).

Considering the areas close to Nordseecluster OWF, five zones are distinguished and marked A to E in Figure 3.21. Table 3.1 presents the mean values and range of the morphological space, depth, median grain size as well as the period of available surveys of these five zones, compared to Nordseecluster OWF. For all areas two surveys, 14 to 26 years apart, are available, allowing for the derivation of the maximum observed bed level difference. Zones D and C extend up to 15 m shallower water depths, while Zones A and B extend up to 4 m deeper water depths compared to Nordseecluster. Lastly, Zones B and C consist of finer sediment sizes (especially towards the east) compared to the sediment sizes at Nordseecluster. Zone D presents coarser material, while Zones A and C are more similar in terms of sediment grain sizes to Nordseecluster OWF.

The morphological space over all zones is on average in the order of a few decimetres. Locally in zone A higher bed level changes (up to 7 m) are observed – potentially linked to human activity in the area. Southwest of Nordseecluster (in Zone D) the patterns in bed level change indicate bed form migration. In this area low long bedforms were observed and described in 3.2.4.1. In Zone C, patterns resembling in orientation the observed sand ridges/ribbons are observed. Those may indicate morphodynamic activity of those features – or may be a by-product of comparing bathymetries of significantly different resolutions.

Apart from these local patterns present in the maximum bed level differences, there is no indication of specific morphodynamic patterns over the 5 considered areas. The maximum differences do not appear to decrease with increasing depth or with increasing sediment sizes over the considered range. The spatial variation in morphological space is rather linked to the extent, resolution and the track line direction of the different surveys. Additionally, the average morphological space estimates are in the range of the measurement errors.

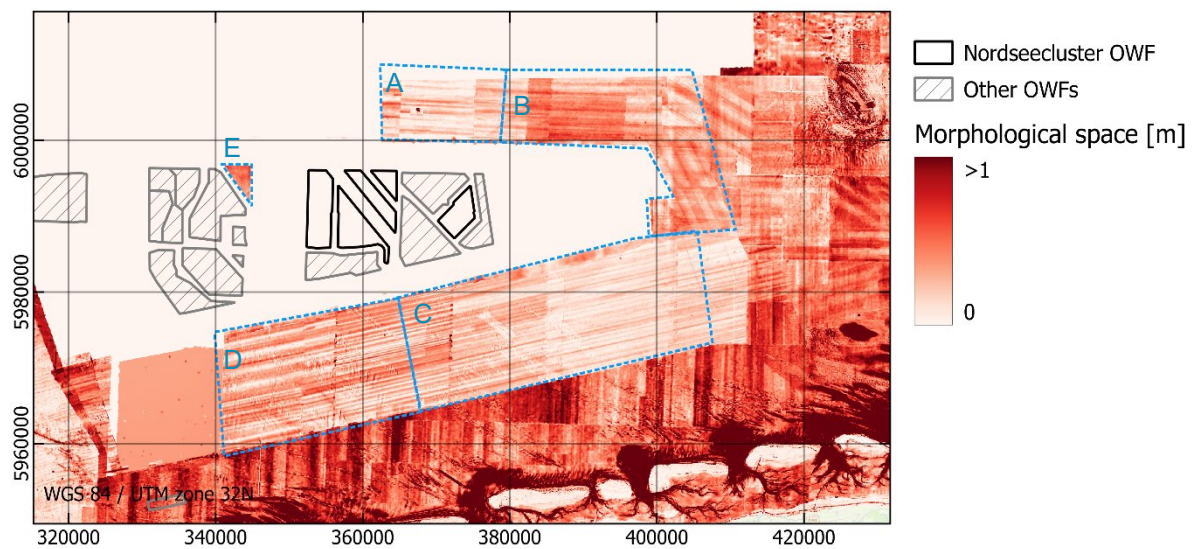


Figure 3.21 Closer view of the morphological space (Bundesamt für Seeschifffahrt und Hydrographie (BSH), 2022) around Nordseecluster OWF. Five zones for which the morphological space is considered separately are marked with the dashed cyan lines.

Table 3.1 Overview of the average morphological space, median grain diameter, depth for each of the five zones marked in Figure 3.21 and Nordseecluster OWF. Minimum and maximum values are presented also for the above characteristics. Lastly, the periods over which the available surveys were used in the calculation of the morphological space are listed.

Zone	Morphological space [m] Mean, (Minimum, Maximum)	D50 [mm] Mean, (Minimum, Maximum)	Depth [m MSL] Mean, (Minimum, Maximum)	Survey period
A	0.14 (0-7)	0.13 (0.03-0.17)	-35.7 (-37.3 -34.1)	1992-2018
B	0.31 (0-2.3)	0.1 (0.05-0.5)	-37.9 (-39.9 -33.45)	1991-2012
C	0.19 (0-1.3)	0.18 (0.09-0.6)	-26.34(-36.3 -20.12)	1987-2001
D	0.30 (0-2.5)	0.28 (0.16-1.05)	-23.3 (-31.1 -15.04)	1987-2003
E	0.37 (0-0.9)	0.16 (0.14-0.18)	-34.4 (-35.9 -31.6)	1991-2010
NSC OWF	Insufficient surveys	0.16 (0.09-0.46)	-33.3 (-35.9 -30.0)	1992

Figure 3.22 and Figure 3.23 present the results of short-term (approximately one year) and long-term (100 years) morphodynamic simulations over the German Bight conducted within the research program AufMod (Heyer et al., 2014). The simulations make use of the functional seabed model developed during the research program and take the whole variability of tides, external surge, river run-off, wind and waves into account (Zeiler et al., 2014). The maps provide insight as to the morphodynamic activity or inactivity of different areas as well as the presence of purely erosive/accretive areas. Figure 3.23 show mainly areas of deposition with a few areas of erosion in the offshore are north of the east Frisian coast. The magnitude of the changes is fairly small (in the order of a few centimetres).

The results of the longer-term simulations, presented in Figure 3.23, show small changes in the offshore part of the German Bight. Heyer et al. (2014) noted in their analysis of the results that morphodynamic evolution below the 20m depth contour line is low to non-existent. They underpin an extensive deposition area, extending from southern Helgoland to the north-

western area of the German Bight, with increasing width in a fan-shape. In this area the simulated deposition over 100 years reaches up to 0.5 m.

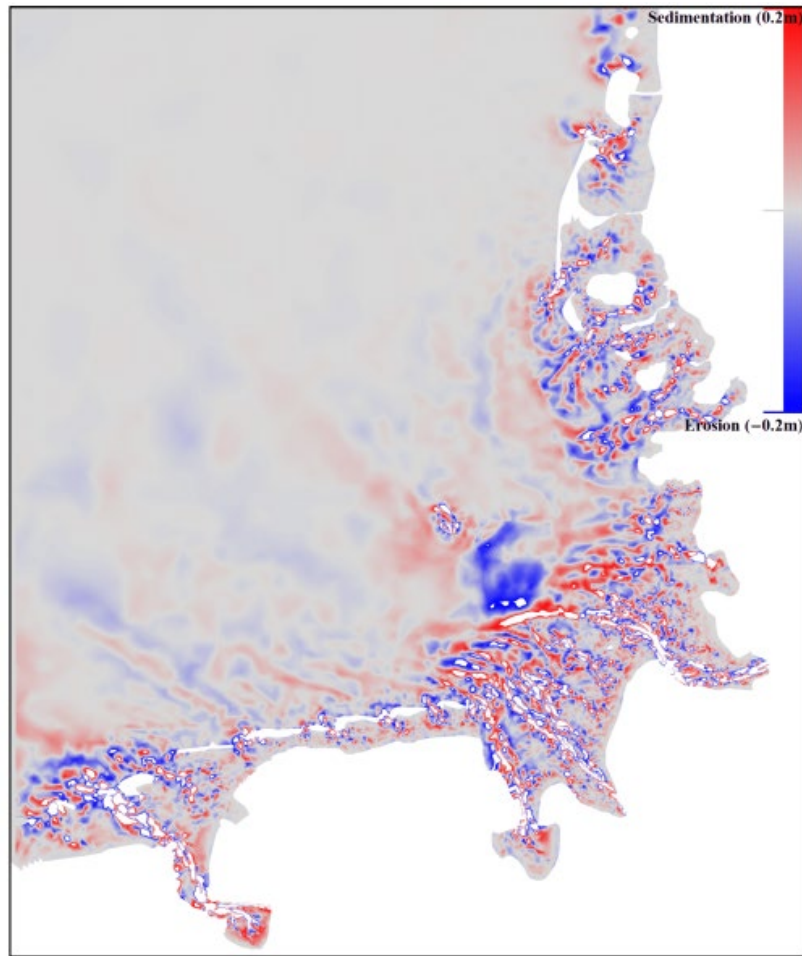


Figure 3.22 Simulated cumulative erosion/sedimentation over 2003 (330 days) over the German Bight (Heyer et al., 2014).

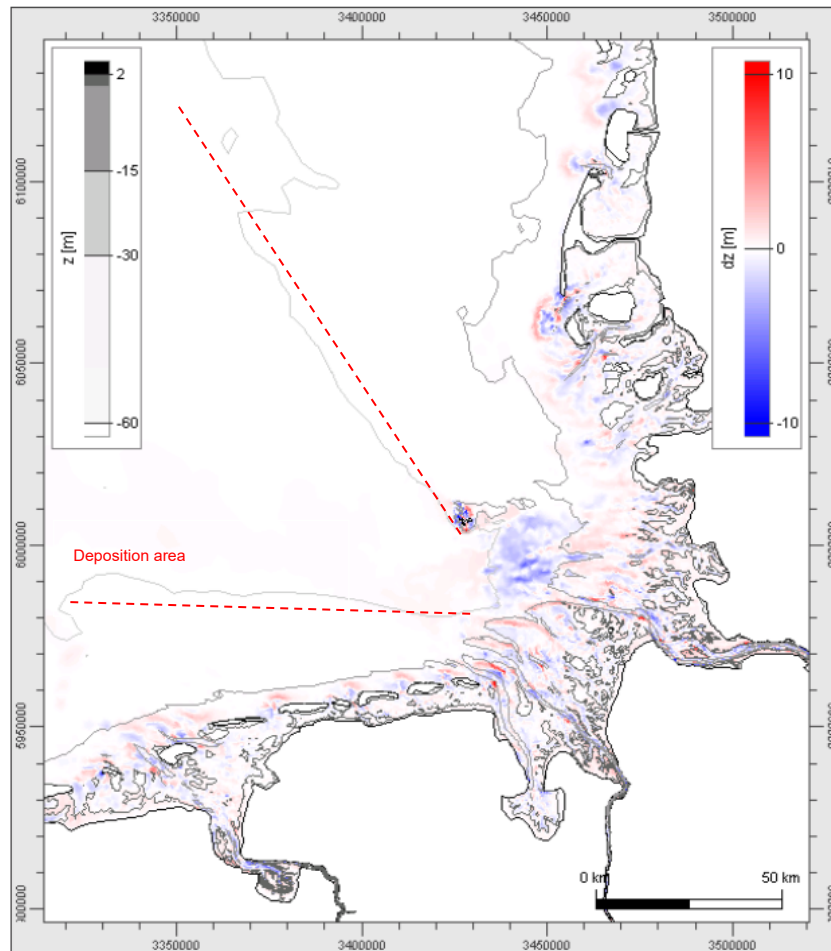


Figure 3.23 Simulated cumulative erosion/sedimentation over 100 years, over the German Bight (Heyer et al., 2014). The approximate extent of the offshore deposition area have been marked with red dashed lines.

Lastly, a data derived analysis of the vertical seabed level change trends over the Dutch continental shelf was conducted by van Dijk et al. (2012). Figure 3.24 presents the derived vertical bed level trends using the methodology described in van Dijk et al. (2012) with updated data sources (until the year 2019), over the eastwards edge of the Dutch Continental Shelf, the area closest to Nordseecluster OWF. As can be seen from the figures, the nearshore zone is particularly dynamic, while the offshore shelf in the depicted area is less dynamic. When considering the areas marked with cyan dashed line polygons the bed level trends are on average lower than ± 0.01 m/year. Slightly higher vertical trends are present at the location of sand ribbons/ridges (at the south marked area) – while the effect of survey errors is also visible. Figure 3.25 shows the number of surveys underlying these results. For the considered areas, the survey number ranges from 2 to 6, which is significantly higher than the 2 surveys used for the derivation of the morphological space around Nordseecluster OWF (Figure 3.21).

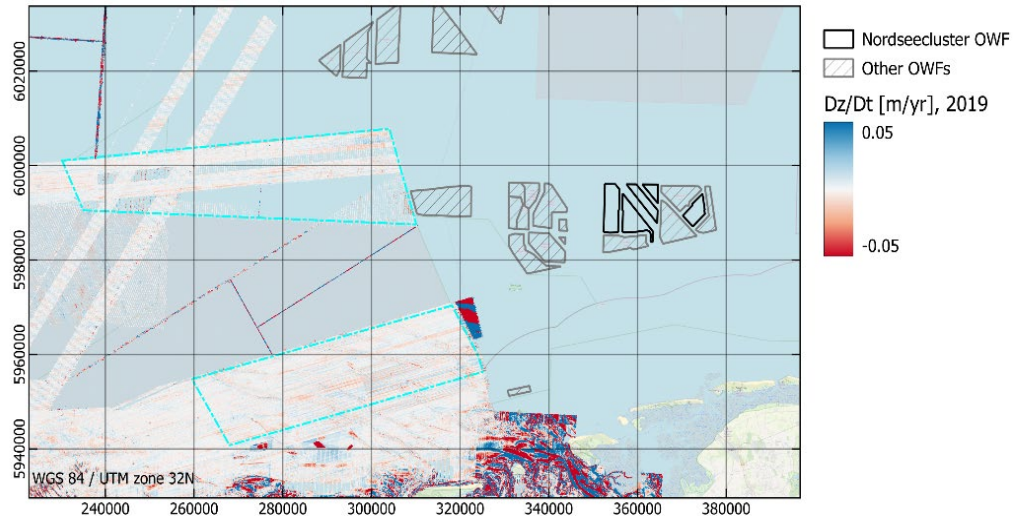


Figure 3.24 Vertical data-derived morphodynamic trends over the east part of the Dutch continental shelf derived according to van Dijk et al. (2012).

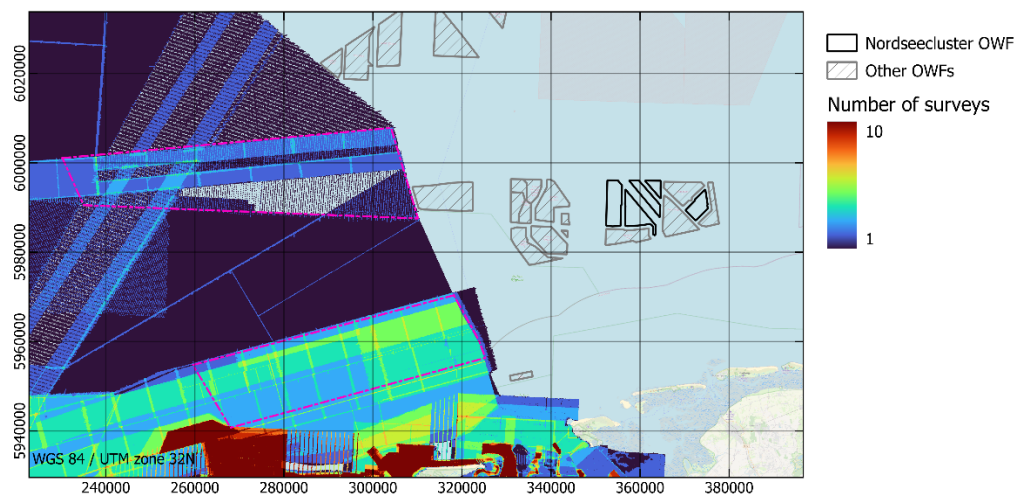


Figure 3.25 Number of available surveys per map point, utilised for the derivation of vertical morphodynamic trends according to van Dijk et al. (2012).

3.3.1.3 Seabed level changes in neighbouring sites

The seabed level changes at two neighbouring areas to Site N-3.8 are considered additionally: Site N-3.7 and Nordsee One OWF.

Nordsee One OWF

Figure 3.26 presents the bed level differences between the four surveys for Nordsee One. Spatially uniform corrections of -0.49 m and 0.15 m have been applied to the 2017 and 2020 surveys respectively for their comparison with the 2018 survey. No correction has been applied for the comparison between the 2020 and 2021 bathymetry. The effect of human interventions is visible in the bed level differences at the WTG locations (including the development of edge scour) and locally along the inter-array cable routes, especially in the 2017- 2018 comparison. Apart from those , bed level changes generally fall below $\pm 0.25\text{m}$ for all compared surveys, with the range slightly reducing for the most recent survey comparison. The standard deviations of the bed level differences are 0.10 m, 0.08 m and 0.06m for the 2017-2018, 2018-2020 and 2020-2021 survey comparisons respectively.

The patterns of remaining bed level differences show alternating behaviour between 2017-2018 and 2018-2020. More specifically, between 2017 and 2018 a bed level increase of a few decimetres is observed over the southwestern part of the OWF and a bed level decrease of a few decimetres is observed over the northeastern part. When comparing the 2018 and corrected 2020 survey, a bed level increase of a few decimetres is observed over the northeastern part of the OWF while a smaller bed level decrease is observed over the remaining area. These patterns could be related to an incorrect tidal correction of the measured bed levels. Alternatively, they could suggest an oscillatory behaviour of the seabed. The specific process behind this potential oscillatory behaviour is not clear from the available data – it might be related with the east to west long-term migration of sediment large-scale, low-height sediment structures or the effect of storms. Conclusions on the potential oscillatory behaviour can be drawn following additional measurements. Such an oscillatory behaviour is however not visible in the bed level differences between 2020 and 2021.

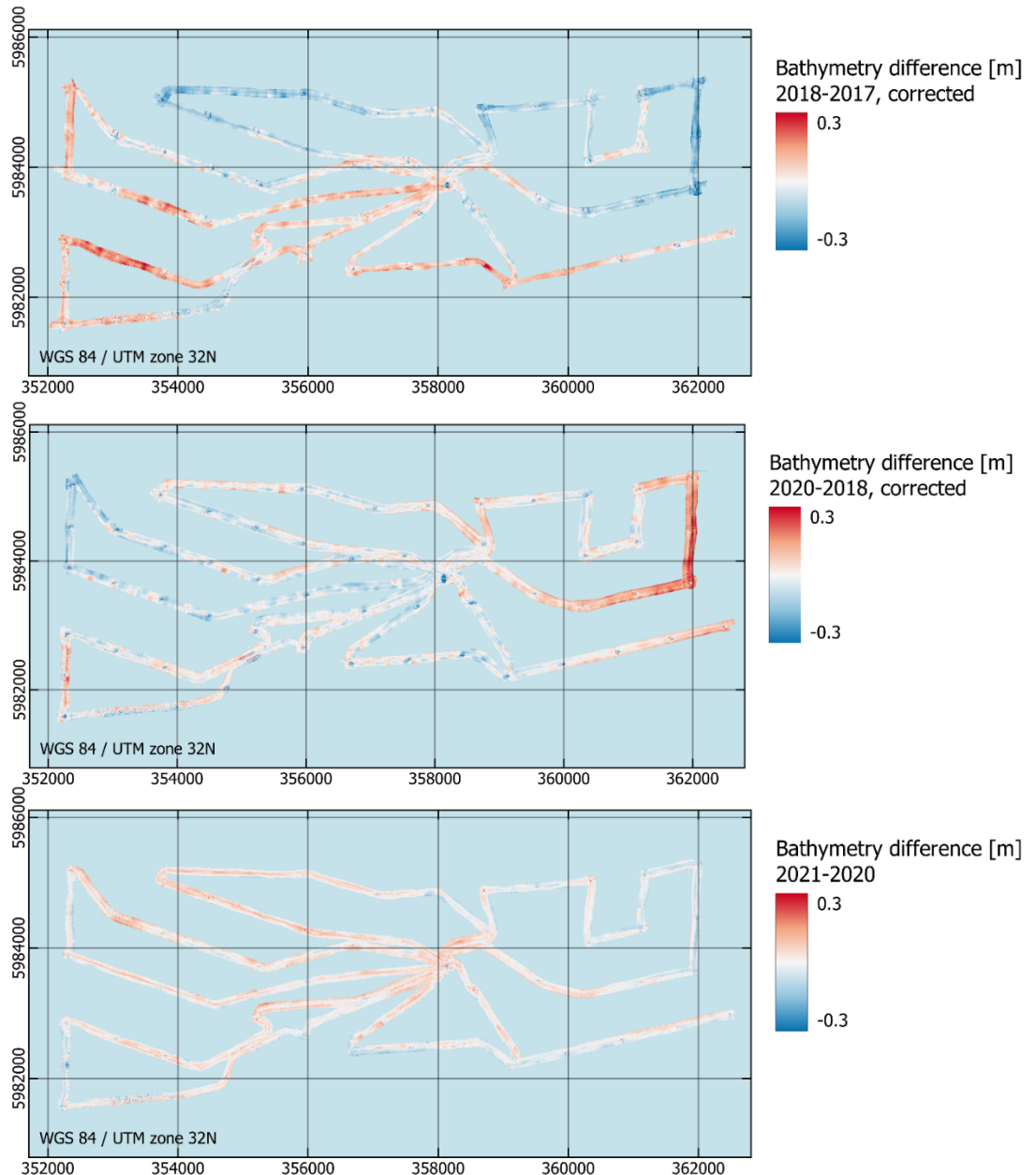


Figure 3.26 Seabed level differences between the 2017 and 2018 surveys (corrected, top panel), 2018 and 2020 surveys (corrected, middle panel) and 2020 and 2021 survey (bottom panel), over the Nordsee One OWF.

Site N-3.7

The analysis of the available surveys over the neighbouring Site N-3.7 (Deltares, 2023) indicated that seabed mobility over this Site is limited as well. After the correction of the bathymetries the remaining seabed changes are estimated at maximum 0.20 m over a period of 3 to 27 years. Similar to Site N-3.8, for this area no clear indications of a persistent seabed level change trend are found. Observed differences fall within the survey inaccuracy bands.

3.4 Estimation of future bed levels

The morphodynamic analysis of the available surveys indicates that over the period 1992 to 2022 the mobility of the seabed in Site N-3.8 is limited. After the correction of the bathymetries (for potential differences in the vertical reference levels), the remaining seabed changes are estimated at maximum 0.12 m over a period of 4 years and 0.46 m over a period of 26 years. It is noted though that the large differences between the 1992 and 2018 datasets are likely connected to large survey inaccuracies of the older dataset. The spatial distribution of the remaining seabed changes suggests that the most extreme values might still be attributed to survey inaccuracies/errors rather than a seabed change trend.

The analysis of the available surveys over the neighbouring Site N-3.7 indicated that seabed mobility over this Site is limited as well. After the correction of the bathymetries the remaining seabed changes are estimated at maximum 0.14 m over a period of 3 years and 0.20 m over a period of 27 years. Similar to Site N-3.8, for this area there are no clear indications of a persistent seabed level change trend.

The recorded bed level changes in the neighbouring Nordsee One OWF are largely dominated by changes related to human interventions (such as the development of edge scour next to WTG locations). The remaining seabed change differences show alternating patterns of erosion/ accretion (within ± 0.25 m) over different parts of the windfarm and different periods. These patterns could suggest an oscillatory behaviour of the seabed, potentially related to migration of large-scale, low height sediment structures or storm impact.

Data driven estimates of morphodynamic activity of the surrounding areas within the German and Dutch continental shelf show limited morphodynamic activity, in the order of 0.25 m over 14 to 26 years. Lastly morphodynamic simulations indicate an overall long-term accretive behaviour of maximum 0.5 m/100 years.

The surveys over Site N-3.7 did not show any evidence of migrating bed forms. Over Site N-3.8 surveys indicated the presence of small scale bedforms (megaripples) of 0.10 m heights (± 0.05 m around a mean seabed). Similar bedforms were observed over NordSee One OWF with their heights increasing to 0.30 m at the southwest edge of the windfarm.

Based on the results of the morphodynamic analysis summarised in the paragraphs above, it is expected that over the lifetime of the windfarm seabed mobility will be limited. It is expected that seabed level changes will occur under the migration of small-scale bedforms. The migration of larger megaripples such as those observed at the southwest of Nordsee One OWF towards Site N-3.8 is considered highly unlikely because of the differences water depths and seabed sediments properties between the sites but cannot be ruled out. It is stressed that this possibility is not included in the predicted seabed levels.

It is also not possible to rule out large-scale seabed mobility based on the current morphodynamic analysis. The limited temporal and spatial spread of the bathymetric surveys and differences in vertical reference levels make it difficult to derive if any large-scale changes occur in the area. Occurrence in Site N-3.8 of the oscillatory behaviour such as that observed in Nordsee One OWF cannot be ruled out. It is expected though that the seabed change would

be smaller than what is observed in Nordsee One, due to the larger water depths in the area. To account for the seabed mobility caused by large-scale morphodynamics a uniform uncertainty of ± 0.20 m (over 30 years) over the full extent of Site N-3.8 is considered based on the review of the data- and literature-derived seabed level differences (see paragraphs above).

It is noted that the observed seabed changes are influenced by data artifacts at the interface between track lines of the multibeam surveys and the differences in vertical reference levels. Moreover, recent multibeam measurements generally have a vertical measurement accuracy of approximately 0.20 m (International Hydrographic Organization, 2020). Most of the observed differences between the 2019 and 2022 bathymetries fall within this uncertainty band.

Together, the uncertainties related to large-scale seabed morphodynamics (± 0.20 m), the uncertainties related to multibeam measurement (± 0.20 m) and the seabed level changes due to small-scale bedforms (± 0.05 m) result in an maximum expected change of ± 0.45 m with respect to the smoothed BSH 2018 bathymetry at any given location over the lifetime of the windfarm (until 2053). It is noted that seabed level changes due to extreme events were not evaluated within this study and are thus not included in the predicted bed levels.

The formulation for the predicted seabed level in any given year of this period is the following:

$$\text{Seabed level}_{2018-2053} = \text{Large-scale seabed level}_{2019} \pm (0.20 \text{ m (large-scale changes)} + 0.20 \text{ m (survey inaccuracy)} + 0.05 \text{ m (small scale bedforms)})$$

The BSH 2018 bathymetry forms basis of the predictions as it is the most recent survey with sufficiently high resolution and full coverage of the site. The bathymetry is smoothed using a to remove the survey artifacts (see Section 0) from the predictions. This formulation is used to determine the lowest seabed level (LSBL) and the highest seabed level (HSBL) over Site N-3.8. The LSBL and HSBL remain constant for the lifetime of the windfarm (up to 2053). The results are provided in GIS data package.

As no specific trend of seabed level change is discerned from the observed bed level differences, the smoothed BSH 2018 bathymetry is determined as the Best Estimate Bed (BEB) level for 2025 (the year of construction). The BEB₂₀₂₅ is on average the best estimate of the 2025 seabed levels. Locally, this level can differ from the actual measured seabed level for that year. It is additionally noted that no uncertainties (e.g., related to survey inaccuracies) have been incorporated in the BEB₂₀₂₅.

3.5 UXO analysis

In this section a UXO analysis is conducted to provide an indication of the lowest seabed levels over the period since World War II relevant to determine possible locations of UXO's. It is assumed that relatively small objects such as Unexploded Ordnances (UXO's) cause no/negligible flow disturbance themselves and therefore will only cause local scour that can result in partial settlement of the object; these objects will, however, not affect the processes responsible for bed form migration or large-scale seabed level changes.

An UXO is expected to never travel upwards and a typical UXO will self-bury to about half its height. It is thus expected that an UXO will most likely stay at the lowest seabed level it has experienced between 1945 and now. Quantification of initial penetration of UXO's into the seabed is not part of the scope of this study. If significant penetration occurred during impact, then at locations that mainly experienced seabed rising the actual vertical level of the UXO's may be overestimated.

In this study estimates of the Lowest and Highest Object Levels, i.e., the minimum and maximum lowest seabed level since the World War II respectively, are calculated based on the results of the morphodynamic analysis. For the Lowest Object Level (LOL), the following components are combined:

1. Lower envelope of the large-scale seabed over the specific period similar to the starting large-scale seabed in case no clear vertical trend is observed);
2. Lower envelope of any migrating bedforms;
3. Downward uncertainty band for the specific period;
4. Top of any non-erodible layer.

For Site N-3.8 the lower envelope of the large-scale seabed over 1945 to 2018 is similar to the 2018 large-scale seabed since no clear vertical trends have been observed. A -0.05 m contribution to account for the presence of small-scale bedforms is considered, similar to the bed level predictions. The downwards uncertainty related to the large-scale seabed morphodynamics is increased to -0.35 m, compared to the respective uncertainty band used in the predictions. This is done to account for the increased time-period (64 years) during which accretion of the seabed (with maximum rate of 0.5 m /100 years) is estimated to have occurred over the area. Additionally, a downward uncertainty band of (-0.20 m) related to multibeam measurement inaccuracy is included. No indications of the presence of non-erodible layers close to the seabed are found. Summing all the above, the Lowest Object Level over N-3.8 is estimated at -0.60 m below the 2018 large-scale bathymetry.

The Highest Object Level is based on the 2018 large-scale bathymetry including an upwards uncertainty of +0.20 m to account for multibeam measurement errors/inaccuracies.

3.6 Risks and possible mitigation measures

A number of different bed level definitions are derived such that the Client may reflect upon these different bed level definitions when considering different design choices.

Morphodynamic activity may pose a threat to foundations and cables if not considered in the design and general wind farm planning. It is beyond the scope of this report to give specific design recommendations, but in the following a few general points of attention are highlighted. The highlighted points should not be taken as exhaustive.

When defining the initial conditions for the design basis, the LSBL and HSBL at the time of foundation installation should be taken into consideration since the seabed in some areas may have changed relative to the most recent 2019 bathymetry. Furthermore, future morphodynamic variations should be considered when estimating the variations which may be observed during the lifetime of the wind farm.

3.6.1 Cables

Within the offshore wind industry currently 70-80% of insurance claims are related to failures of cables (<https://www.deltares.nl/en/news/launch-major-joint-industry-project-reliable-offshore-cables/>). On average in Europe one export cable and about 10 inter-array cables fail every year. Cable failures pose one of the highest risks as it can blackout an entire wind farm. In addition, cable monitoring and repair require expensive marine operations. One of the causes of cable failures is morphodynamic activity such as sand wave migration. Typical failure mechanisms are:

- Manufacturing faults;
- Insufficient cable burial depth;
- Overheating;
- Internal stresses;

- Free spanning;
- Dragging anchors or fishnets, dropped objects.

With lowering seabed (due to large-scale seabed level changes or bed form migration), a cable may experience significant seabed lowering, which may make the cable vulnerable to anchors or other threats. On the other hand, with rising seabed levels a cable may experience a significant increase in the burial depth, which locally may cause temperature increases around the cable. Depending of the specifications of the cable and environmental requirements, this may be a problem.

Cables may experience a local stress build-up due to an uneven strain under spatially varying bed level changes. When combined with e.g. thermal stresses this may become critical. It is well known that cables exposed on the seafloor may experience local scour, which in some cases may be sufficient to undermine the cable, causing a free span. A free span of a cable may, besides a local stress build up, also experience vortex induced vibrations.

3.6.2 Pile foundations

Seabed level changes may also pose a problem to the foundations of the wind turbines or sub-stations. Large seabed changes may cause problems with respect to:

- Geotechnical stability due to reduced support
- Stability of scour protection
- Change in eigen-frequencies

If a foundation is installed in an area of lowering seabed levels (due to large-scale seabed level changes or bed form migration) it may experience a significant lowering, which combined with e.g. scour may cause insufficient geotechnical bearing capacity due to reduced support from the surrounding soil. One way to prevent this is installation of scour protection systems, however, if the scour protection is not sufficiently flexible and able to adjust to the seabed variations it may become unstable and in worst case fail to protect the foundation. Therefore, locations with large expected seabed lowering are best avoided.

As the fixation level of the pile changes due to morphodynamic activity, the dynamics of the combined system including foundation and tower may change. In the worst case the natural frequency of the system changes which may lead to an undesired amplification of harmonic loading.

3.6.3 Summary

In the two paragraphs above, potential risks are identified for cables and pile foundations. Note that only risks related to morphodynamic and mobile seabed are summarized in Table 6.1; other potential risks are not addressed. Besides, potential risks related to the usage of a jackup platform during the installation of wind turbines are also added to the table. This section is indicative only and is not intended to be complete or comprehensive.

Morphodynamic risks are related to large-scale seabed variations (due to natural processes, unrelated to the presence of man-made structures); risks related to a mobile seabed here refer to local interaction between the hydrodynamics, the structure and the mobile seabed. As can be seen from the table below, potential risks can be mitigated by either a careful selection of the location with respect to expected seabed lowering or by taking appropriate mitigation measures or by a combination of both.

Table 3.2 Overview of potential risks related to morphodynamic and mobile seabed for cables as well as pile foundations (and a jackup platform if used for the installation); this table is indicative only and not intended to be complete or comprehensive.

Structure type	Potential risks related to morphodynamics of the seabed	Potential risks related to mobile seabed (sediments)
Cables	Negligible risk in areas with a stable seabed; low risk in areas with a (slightly) rising seabed if thermal characteristics of the cable are taken into account in cable design; significant risk on cable exposure in areas with a lowering seabed and a small initial cable burial depth. For Site N-3.8 uniform bed level changes of maximum ± 0.45 m within the period 2018-2053 are predicted. Assuming that the predicted seabed mobility is considered in the design and placement of the cables, potential risks related to seabed morphodynamics on the cables are assessed low.	As long as the cable is buried sufficiently deep (for other potential threats such as anchor dragging, dropped objects etc.) the risks are low to negligible. Special attention should be given to the areas just around the scour protections of the wind turbine foundations, where due to edge scour the cables may become exposed after some years. Also, cable crossings require special attention.
Pile Foundations	Significant risk for change in eigen-frequency if piles are installed at unfavourable locations and morphodynamics are not taken into account in the structure and/or scour protection design. When installed at carefully selected locations the risks can be low to negligible. For Site N-3.8 uniform bed level changes of maximum ± 0.45 m within the period 2018-2053 are predicted. Assuming that the predicted seabed mobility is considered in the design and placement of the pile foundations, potential risks related to seabed morphodynamics on the pile foundations are assessed low.	Scour around the foundation might change the eigen-frequency of the pile. Pile foundations can potentially be designed for the expected scour depth, but a scour protection might be more cost-efficient, especially for larger turbines and larger pile diameters.
Jackup Platform (if used)	Negligible risk due to limited duration of jack-up operations (relative to the timescale of morphodynamic processes)	Low risk for short-term operations (of a few days), significant risk for longer operations (weeks to months) depending on the leg and spud can type and penetration depth. Scour protection might be required also for temporary operations.

4 Conclusions and recommendations

4.1 Conclusions

Deltares has carried out a seabed mobility study for Site N-3.8 of Nordseecluster OWF, located approximately 55 km off the coast of Germany in the North Sea. Site N-3.8 is one of the four sites (N-3.5, N-3.6, N-3.7 and N-3.8), that are being developed as a single combined project by Nordsee Two GmbH and Nordseecluster B GmbH. This report describes the methods and results of the seabed mobility analysis within and around one of the sites, Site N-3.8. The objectives of the seabed mobility study are to assess the morphodynamics and provide an estimation of the future seabed levels in the area of interest. To this end a data-driven analysis is performed that consists of five parts: overview of the characteristics in Site N-3.8 morphodynamic analysis and estimation of future bed levels, Unexploded Ordnance (UXO) analysis and a description of possible risks related to the predicted morphodynamic changes and possible mitigation measures.

The hydrodynamic and wave data combined with a literature review of sediment transport in the area show a west to east directed sediment transport pattern driven by tide and meteorological forcing. The geotechnical and geophysical data illustrated that the substrate in and around Site N-3.8 is characterised by fine sands. The morphodynamic analysis showed that over the period 1992 to 2022 the mobility of the seabed in Site N-3.8 is limited. Nevertheless, seabed mobility cannot be fully excluded due to the limited temporal and spatial coverage of the bathymetric surveys and uncertainties related to the measured bathymetric datasets.

To account for seabed changes caused by small-scale dynamics (i.e., migration of megaripples) and large-scale dynamics (i.e., regional seabed changes) we thus consider a uniform uncertainty of ± 0.25 m over the full extent of Site N3.8. Together with the uncertainties related to multibeam measurement (± 0.20 m) a maximum change of ± 0.45 m with respect to the smoothed BSH 2018 bathymetry is expected at any given location over the lifetime of the windfarm (30 years until 2053).

4.2 Recommendations

Although it is concluded that there is little potential for seabed mobility in this region, there was limited availability of bathymetry to confirm this further. It is thus advised to further expand the current available bathymetry datasets over time (during installation), to confirm the anticipated behaviour as described in this report. An additional bathymetry dataset provides further confirmation on the offset in the 2018-2022 bathymetries, a third point (or fourth if considering the older 1992 data) in time to increase the temporal and spatial resolution of the available data and a starting point during installation of the wind farm. It is stressed that a full area survey is advised instead of measurements at the locations of the foundations.

During wind farm operation, it is advised to monitor the seabed around each foundation periodically (e.g., every two years) or following extreme events (e.g. after the occurrence of an extreme event). This way, potential risks related to seabed mobility can be identified and mitigated beforehand.

In addition, it is advised to address the potential effects of wind farm related infrastructure in relation to the observed trends in this study. Although the observed autonomous bed level changes are not large, there is a clear tide and meteo-driven, net sediment transport direction which could lead to sediment infill of e.g., a cable trench.

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A Appendix A

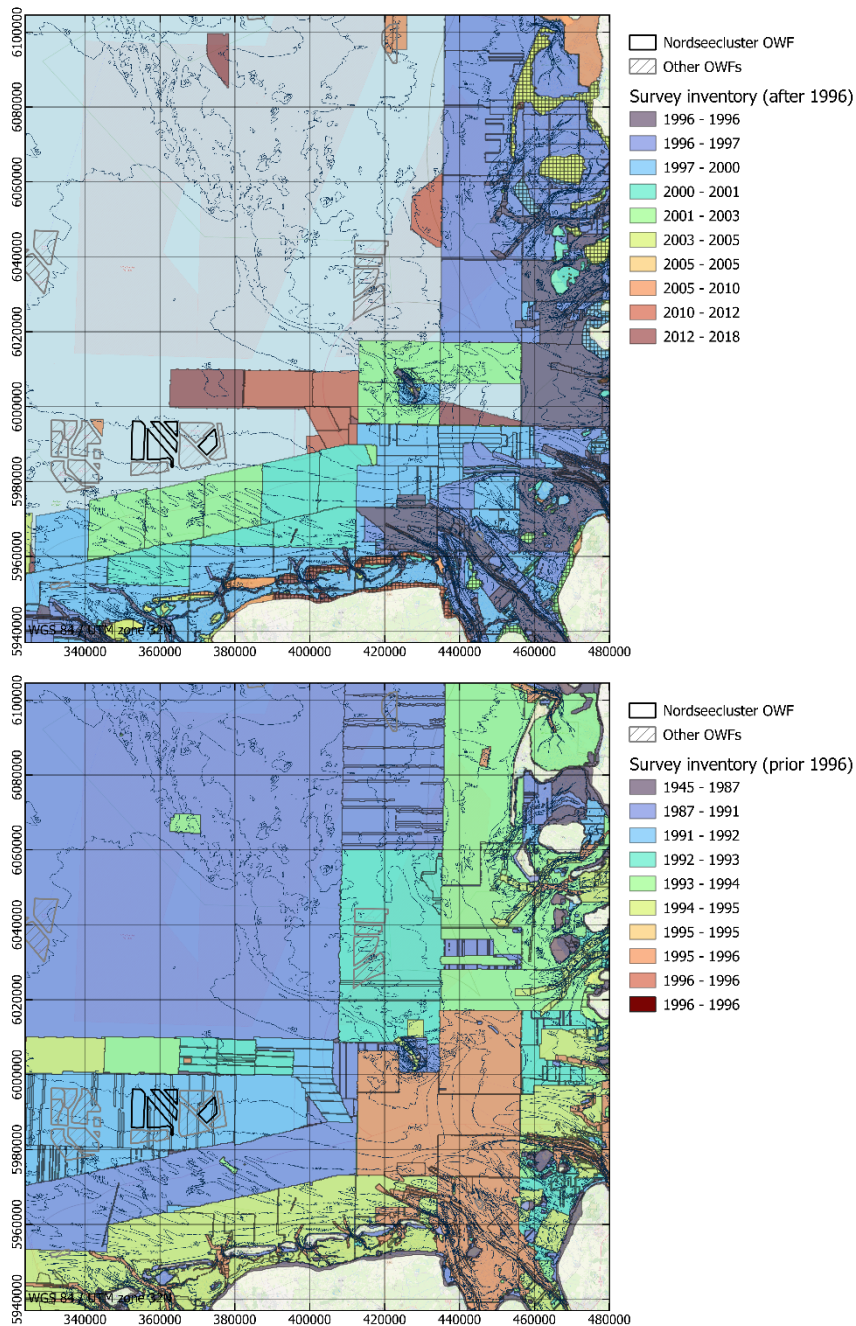


Figure A.1 Survey extents and year of acquisition of surveys after 1996 (top panel) and before 1996 (bottom panel) over the German Bight. The surveys were used in the assessment of the morphological space (Bundesamt für Seeschifffahrt und Hydrographie (BSH), 2022).

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